

SpokenUS: A Spoken User Simulator for Task-Oriented Dialogue

Anonymous ACL submission

Abstract

Robust task-oriented spoken dialogue agents require exposure to the full diversity of how people interact through speech. Building spoken user simulators that address this requires large-scale spoken task-oriented dialogue (TOD) data encompassing spoken user behaviors, yet existing datasets are limited in scale and domain coverage, with no systematic pipeline for augmenting them. To address this, we introduce **SpokenTOD**, a spoken TOD dataset of 52,390 dialogues and 1,034 hours of speech augmented with four spoken user behaviors—cross-turn slots, barge-in, disfluency, and emotional prosody—across diverse speakers and domains. Building on SpokenTOD, we present **SpokenUS**, a spoken user simulator grounded in TOD with a dedicated architecture for barge-in. SpokenUS achieves comparable goal coverage to significantly larger models while substantially outperforming all baselines in Human MOS, disclosing slot values gradually across the dialogue as humans do rather than front-loading them. Further analysis confirms that SpokenUS’s spoken behaviors pose meaningful challenges to downstream agents, making it a practical tool for training and evaluating more robust spoken dialogue systems.¹

1 Introduction

When interacting with dialogue agents, people often produce disfluencies (Shriberg, 1994), make interruptions (Khouzaimi et al., 2015), and express emotion (Schröder, 2001) rather than following clean, scripted exchanges, and such spontaneous behaviors are known to degrade the performance of deployed dialogue agents (Liu et al., 2025). Building spoken dialogue agents robust enough to handle this diversity is critical for real-world task-oriented deployments such as smart homes (Amazon, 2026; Seo et al., 2026) and voice assistants (Apple, 2024),

where systems must reliably assist users in accomplishing concrete goals. Yet, collecting sufficient spoken dialogue data for training such systems is prohibitively expensive.

User simulation offers a scalable alternative (Eckert et al., 1997), where a user simulator is provided with a user goal (i.e., requests and all slot values) and interacts with an agent turn by turn to fulfill it. However, existing simulators operate in the text modality (Sekulic et al., 2024; Shim et al., 2026) and fundamentally cannot capture the spontaneous behaviors that define spoken interaction. Although attaching a TTS module could partially address this by introducing spoken behaviors like emotional prosody, real-time interruption behaviors such as barge-in cannot be simulated within a sequential ASR-LLM-TTS pipeline.

End-to-end omni models (Xu et al., 2025a,b; Zeng et al., 2024; Tong et al., 2025) offer a more promising direction by processing and generating speech directly, and in principle support real-time interaction. However, without grounding in task-oriented dialogue (TOD) knowledge, such models struggle to consistently deliver slot values and requests as stated in the provided goal statement across turns. Moreover, as these models are designed to act as dialogue assistants rather than to simulate users, their architectures provide no support for proactive turn-taking behaviors such as barge-in, which a user simulator must exhibit to test how spoken assistants handle interruptions. Addressing these limitations is further complicated by data scarcity: existing spoken TOD datasets (Henderson et al., 2014; Kim et al., 2021; Si et al., 2023) are limited in scale and domain coverage, and no systematic pipeline exists for augmenting text-based TOD corpora with diverse spoken user behaviors, making it infeasible to train spoken user simulators grounded in both task-oriented domain knowledge and realistic spoken behaviors.

To address these limitations, we introduce **Spo-**

¹We will release our code upon publication.

Category	Type	Definition	Example or Details
Cross-Turn Slots		Slot values that span multiple turns.	Phone numbers, Email addresses.
Barge-in	Error Recovery	Correct an incorrect system response.	S : “For two peo...” U : “No, it’s three.”
	Clarification	Request details on an unclear response.	S : “It’s nearby...” U : “Which address?”
	Efficiency	Move on when sufficiently informed.	S : “Successfully ...” U : “Thank you!”
Disfluency	Filled Pause	Non-lexical vocalized hesitation.	"[FP] uh, we should go there."
	Disc. Marker	Lexical filler with no semantic content.	"[DM] well, I guess so."
	Edit	Editing cue signaling a self-repair.	"reserve, [EDIT] I mean, at four."
	Restart	Abandoned utterance with a fresh start.	"I went to the- [RST] I went to the store."
	Repetition	Immediate repetition without change.	"I mean, [REP] I mean I don’t know."
	Correction	Substitution altering the intended meaning.	"left, [COR] no, right after the bank."
Emotion		Emotional reactions of users.	Neutral, Satisfied, Dissatisfied, etc.

Table 1: A taxonomy of spoken user behaviors in the SpokenTOD dataset. **S** denotes System and **U** denotes User.

kenTOD, a large-scale spoken TOD dataset that augments existing text-based corpora with four spoken user behaviors: cross-turn slots, barge-in, disfluency, and emotion-aware prosody (Table 1), spanning 52,390 dialogues and 1,034 hours of speech across diverse speakers and domains. Building on SpokenTOD, we present **SpokenUS**, a spoken user simulator grounded in TOD, supporting diverse spoken user behaviors, and enabling turn-taking through a dedicated interruption mechanism.

We evaluate SpokenUS against five end-to-end omni models across goal coverage, Human MOS, and intelligibility. SpokenUS achieves comparable goal coverage to significantly larger models, while informing slot values gradually across the dialogue as humans do, rather than front-loading them as omni models tend to do. It further outperforms all baselines in Human MOS (4.06 vs. 3.18) despite training on orders of magnitude less audio data (~1K vs. ~20M hours). We also demonstrate that SpokenUS’s spoken behaviors pose a meaningful challenge to agents, yielding larger performance drops than existing omni models under a cascaded ASR pipeline. Ultimately, we present SpokenUS as, to our knowledge, the first LLM-based spoken user simulator suited for evaluating (and potentially training) spoken dialogue systems; alongside the release of SpokenTOD, we call for future work to build upon and advance this research direction.

Our contributions are as follows:

- We present **SpokenTOD**, a large-scale spoken TOD dataset of 52,390 dialogues and 1,034 hours of speech, augmented with cross-

turn slots, barge-in, disfluency, and emotional prosody across diverse speakers and domains.

- We present **SpokenUS**, a spoken user simulator grounded in TOD, supporting diverse spoken user behaviors and enabling turn-taking via a dedicated interruption mechanism.
- We show that SpokenUS generates realistic spoken behaviors that pose a meaningful challenge to agents, making it a practical tool for training more robust spoken dialogue systems.

2 SpokenTOD

To construct a spoken TOD dataset that captures realistic user behaviors, we augment existing text-based TOD datasets with these behaviors and synthesize the dialogues into speech rather than collecting spoken dialogues from scratch (Figure 1). This approach offers two key advantages. First, by building on human-annotated corpora, we preserve the conversational flow that emerges from real human interactions, and rich annotations such as dialogue states and system acts. Second, our augmentation pipeline is dataset-agnostic: it can be layered on top of any existing TOD dataset construction pipeline, enabling generation of high-quality spoken TOD data across new domains. Moreover, it is compatible with any TTS backbone, allowing the pipeline to benefit from future advances in speech synthesis.

We build upon four widely used datasets: ABCD (Chen et al., 2021), EmoWOZ (emotion-labeled MultiWOZ) (Feng et al., 2022), SGD (Rastogi et al., 2020), and TaskMaster (Byrne et al., 2019).

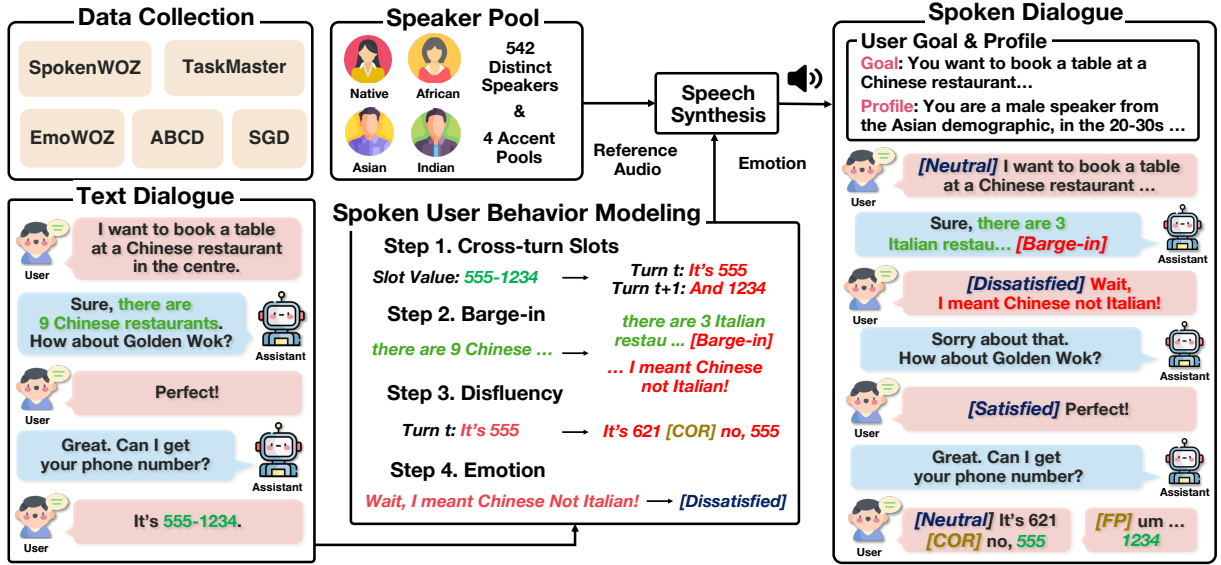


Figure 1: SpokenTOD Construction Pipeline.

2.1 Spoken User Behaviors

In this section, we detail how we augment TOD datasets with spoken user behaviors: cross-turn slots, barge-in, disfluencies, and emotion (Table 1). For all LLM-based augmentation and annotation steps, we use Qwen3-32B (Yang et al., 2025).

Cross-Turn Slots. Humans often provide complex slot values such as phone numbers and email addresses across multiple turns rather than stating the full value at once (McInnes and Attwater, 2004; Zhang et al., 2022), reflecting listeners’ limited memory spans: each chunk is confirmed before the next is presented (Clark and Brennan, 1991). Speakers may also revise previously provided slot values in subsequent turns. Recent work (Si et al., 2023) has identified these behaviors collectively as *cross-turn slots*, a key challenge for spoken dialogue agents that must collect, verify, and revise partial slot values across turns. To model this, we segment all long alphanumeric slot values into natural chunks (Step 1 in Figure 1) and additionally augment user-initiated self-corrections (e.g., Turn t: *table for two* → Turn t+1: *Actually, three*). Further details are provided in Appendix A.3.

Barge-in. In natural spoken dialogue, speakers do not simply alternate complete utterances; they continuously monitor each other’s speech and make real-time decisions about when to take or yield the floor (Sacks et al., 1974; Clark, 1996). We augment our dataset with *barge-in* (Khouzaimi et al., 2015), where the user interrupts the agent to correct

errors (*error recovery*), request clarification (*clarification*), or skip ahead with brief backchannel-like acknowledgments when sufficient information has been conveyed (*efficiency*). Table 1 summarizes the taxonomy. Since existing TOD datasets do not contain barge-ins, we modify the dialogues to insert natural barge-ins as follows. We first sample 25% of user turns as barge-in candidates (Witt, 2015). After each candidate is judged contextually appropriate by the LLM, we insert a new agent utterance as an erroneous turn truncated mid-sentence, followed by the user’s interruption. This barge-in is then naturally followed by the original agent response, which serves as a logical recovery response (Step 2 in Figure 1). We validate the contextual appropriateness of LLM-placed barge-ins through human evaluation, with 83% rated as appropriate. We further confirm that these barge-ins are naturally realized in the synthesized audio (C.10). Detailed augmentation rules, prompt templates and validation results are provided in Appendix A.4.

Disfluency. Disfluency reflects the natural imperfections of real-time speech production, manifesting as hesitations, repetitions, and self-repairs (Shriberg, 1994). Our augmentation follows two stages. First, we select which utterances become disfluent using the length-dependent probability model of Shriberg (1996), where longer utterances are more likely to contain disfluencies. Second, we inject disfluencies from six categories, as shown in Table 1. Unlike previous work that uniformly inserts disfluencies via rules (Yang et al., 2020;

Dataset Overview		Behavior	
# Dialogues	52,390	# Cross-turns	11,045
# Utterances	1,208,554	# Barge-ins	79,141
Avg. Words	9.95	# Disfluencies	118,377
# Speakers	542	# Emotions	511,468
Total Duration	1,034 hrs	Total	720,031

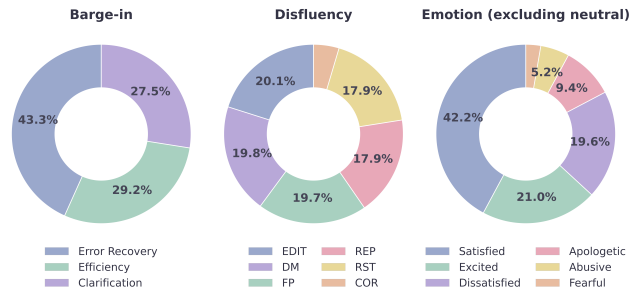


Table 2: Overall statistics of the SpokenTOD.

Passali et al., 2022) or LLM paraphrasing (Marie, 2023), we concentrate insertion in a local window around slot values (Step 3 in Figure 1), as speakers are known to be more disfluent when producing precise information such as names and phone numbers (Corley et al., 2007; Arnold, 2013; Sen, 2020). Further details are provided in Appendix A.5.

Emotion. Beyond the behavioral patterns discussed above, realistic spoken dialogue also conveys speakers’ emotional states. To support emotion-conditioned speech synthesis, we annotate each user utterance with an emotion label using the LLM (Step 4 in Figure 1), following the EmoWOZ label set grounded in the Ortony–Clore–Collins (OCC) model (Ortony et al., 2022): NEUTRAL, FEARFUL, DISSATISFIED, APOLOGETIC, ABUSIVE, EXCITED, and SATISFIED. Agent utterances are all labeled as NEUTRAL, reflecting the expectation that deployed AI systems communicate in a neutral manner (Han et al., 2023). Annotation details and label reliability are in Appendix A.6.

2.2 Speech Synthesis

We synthesize the augmented dialogues into speech using Qwen3-TTS (Hu et al., 2026), controlling *speech style* via emotion labels (§2.1) and *speaker identity* via sampled audio from a speaker pool.

Emotion Conditioning. We control speech style in Qwen3-TTS via instruction prompts that specify the target emotion for each turn. Specifically, we map each emotion category to a set of speaking style keywords (e.g., DISSATISFIED → {*angry*, *contempt*, *disgusted*}) drawn from CosyVoice3’s training data (Du et al., 2025), as both models share overlapping pretraining data. At each turn, we randomly sample one keyword from the corresponding set to introduce variation even within the same emotion label. Full construction details and the keyword list are provided in Appendix A.8.

Speaker Diversity. To simulate a diverse range of real-world users, we sample reference audio clips from the Speech Accent Archive (Weinberger and Kunath, 2011), applying population-weighted sampling across four accent groups with stratification for age and gender balance, resulting in a pool of 542 reference speakers. For assistant utterances, we fix a pool of 10 *Native*-accent speakers to reflect the consistent voice identity expected of deployed AI systems (Han et al., 2023). Further details are provided in Appendix A.9.

2.3 Dataset Statistics and Validation

To complement our synthesized data with authentic human speech, we additionally include SpokenWOZ (Si et al., 2023) as human-recorded samples for both training and evaluation. The resulting SpokenTOD dataset comprises a total of 52,390 dialogues and 1,034 hours of speech. As shown in Table 2, the augmentation pipeline produces 11,045 cross-turn slots, 79,141 barge-ins, 118,377 disfluencies and 511,468 emotion annotations across 720,031 total behaviors, with the distribution of each behavior type illustrated in the pie charts. To validate speech intelligibility, we measure Word Error Rate (WER) on 500 randomly sampled dialogues using Whisper-large-v3 (Radford et al., 2023), achieving an overall WER of 4.69%. A three-way human evaluation (human recordings, Qwen3-TTS without augmentation, full SpokenTOD pipeline) further confirms that SpokenTOD audio approaches human-recorded quality (Naturalness 3.88 vs. 3.95) and our augmentation substantially improves spoken behavior realism over raw TTS (Disfluency 2.97 → 3.76, Emotion 3.59 → 4.33). Further details are provided in Appendix A.10.

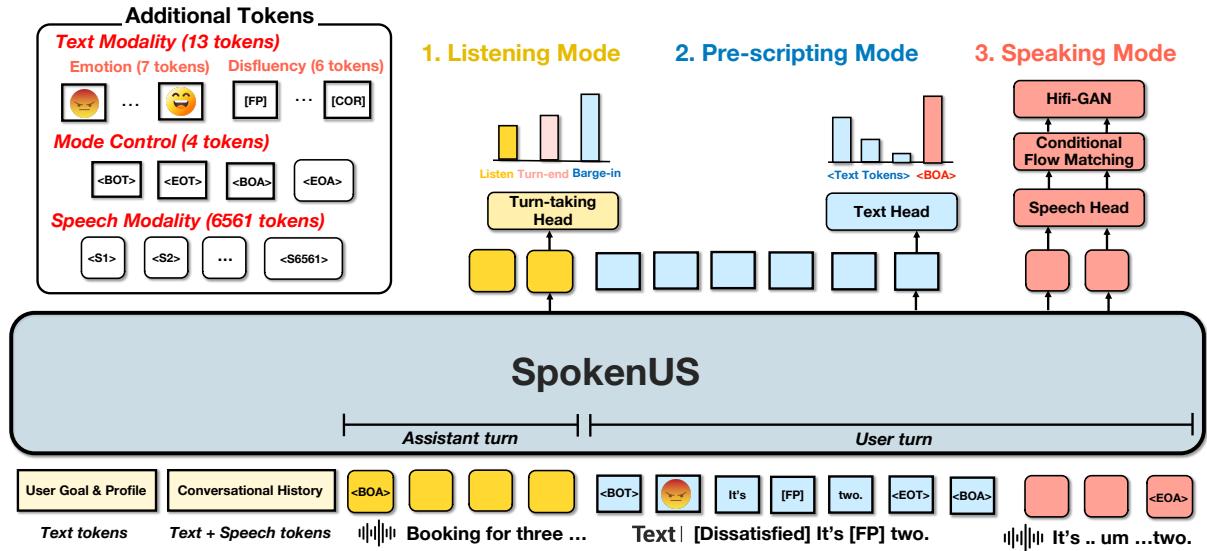


Figure 2: Overview of SpokenUS. The model processes streaming assistant speech in Listening Mode, determines barge-in timing through a turn-taking head, then generates responses through Pre-scripting and Speaking Modes.

3 SpokenUS

3.1 Framework Overview

SpokenUS is a half-duplex spoken user simulator that pursues a given user goal through multi-turn exchanges, modeling user-side turn-taking while expressing realistic spoken user behaviors. As illustrated in Figure 2, it takes the user’s goal and profile, the conversational history as interleaved text and speech tokens, and the current assistant speech, provided either as a complete utterance or as a streaming input. SpokenUS operates in three sequential modes: *Listening Mode*, which monitors incoming assistant speech to determine when to speak; *Pre-scripting Mode*, which generates a transcript before speech is produced; and *Speaking Mode*, which synthesizes the transcript into speech.

3.2 Operating Modes

Listening Mode. The turn-taking head continuously monitors incoming assistant speech to determine when the user should speak. Since audio arrives as a stream of tokens (Noroozi et al., 2024), the turn-taking head produces a prediction at each token and aggregates them by computing a weighted sum of per-token probabilities to avoid false alarms, with linearly increasing weights assigned to more recent tokens. A *listen* prediction allows streaming to continue. A *barge-in* prediction triggers an immediate interruption, while a *turn-end* prediction indicates that the assistant has finished speaking. In either case, the model tran-

sitions to Pre-scripting Mode. When a complete assistant utterance is provided as a single audio file rather than a stream, Listening Mode is skipped.

Pre-scripting Mode. Upon entering this mode, the text head generates a structured transcript delimited by <BOT> and <EOT>. As illustrated in Figure 2, the transcript begins with an emotion token (e.g., *[Dissatisfied]*), followed by the intended utterance with interleaved disfluency tokens (e.g., *[FP] two.*). This explicit transcript guides speech generation in Speaking Mode while making the simulator’s behavior transparent for evaluation.

Speaking Mode. The <EOT> token signals the transition to Speaking Mode, where the speech head autoregressively generates discrete speech tokens conditioned on the transcript, delimited by <BOA> and <EOA>. Since SpokenTOD is constructed with this same structure, emotion and disfluency markers are faithfully realized as prosodic and hesitation patterns. As illustrated in Figure 2, the generated tokens are decoded to audio via a Conditional Flow Matching (CFM) model and HiFi-GAN (Kong et al., 2020) vocoder, conditioned on a reference audio clip drawn from the diverse speaker pool constructed in SpokenTOD (§2.2), enabling SpokenUS to simulate a wide range of user voices. Further details on mode transition are provided in Appendix B.1.

Model	Audio (hrs)	Goal Cov.		Human MOS				Intell.	Speaker Sim.	
		GA $\uparrow$	SMR $\uparrow$	N $\uparrow$	C $\uparrow$	S $\uparrow$	Avg. $\uparrow$	WER $\downarrow$	SIM _{0,i}	SIM _{i-1,i}
Human	–	1.00	1.00	3.58	3.64	3.80	3.67	–	0.84 $\pm$.07	0.88 $\pm$.04
<i>Larger models</i>										
Qwen2.5-Omni-7B	~3M	0.80	0.97	2.86	3.29	2.94	3.03	10.53	0.97 $\pm$.03	0.96 $\pm$.04
GLM-4-Voice-9B	~700K	0.33	0.79	2.51	2.06	2.38	2.32	15.33	0.94 $\pm$.04	0.96 $\pm$.03
Qwen3-Omni-30B-A3B	~20M	0.93	0.99	3.03	3.61	2.91	3.18	15.05	0.95 $\pm$.03	0.95 $\pm$.04
<i>Comparable-sized models</i>										
Qwen2.5-Omni-3B	~3M	0.36	0.93	1.97	2.77	2.28	2.34	15.09	0.90 $\pm$.04	0.92 $\pm$.03
InteractiveOmni-4B	~465K	0.59	0.89	2.83	2.44	2.62	2.63	15.68	0.95 $\pm$.03	0.94 $\pm$.04
SpokenUS (3B)	~1K	0.82	0.99	3.92	4.14	4.11	4.06	11.36	0.93 $\pm$.02	0.92 $\pm$.03

Table 3: Main evaluation results. The best result in each group is highlighted in **bold**.

3.3 Training

SpokenUS is trained with a multi-task objective combining losses from all three heads:

$$\mathcal{L} = \mathcal{L}_{\text{text}} + \mathcal{L}_{\text{speech}} + \mathcal{L}_{\text{turn-taking}} \quad (1)$$

where $\mathcal{L}_{\text{text}}$ and $\mathcal{L}_{\text{speech}}$ are cross-entropy losses for text and speech token generation respectively, and $\mathcal{L}_{\text{turn-taking}}$ is the cross-entropy loss for turn-taking decisions over three classes (listen, barge-in, and turn-end). For turn-taking supervision, all tokens within an assistant turn are labeled *listen* except for the final 6 tokens, which are labeled *turn-end* or *barge-in* depending on the turn type. The 6-token window corresponds to approximately 240ms at 25 Hz, reflecting the average human reaction time when taking a turn (Stivers et al., 2009). Speech token targets are obtained by encoding SpokenTOD audio with the CosyVoice3 tokenizer (Du et al., 2025), which produces 6,561 discrete tokens via Finite Scalar Quantization (Mentzer et al., 2024).

4 Experimental Setup

4.1 Implementation Details

SpokenUS is initialized from Qwen2.5-3B (Qwen et al., 2025) and fine-tuned in two stages. We first augment the vocabulary with disfluency and emotion special tokens (Table 9). In Stage 1, the model is trained on text-only SpokenTOD for 3 epochs to acquire TOD knowledge. In Stage 2, all three heads are jointly trained for 12k steps using the multi-task objective in Section 3.3, supervised on user turns only. The CFM and HiFi-GAN are trained exclusively on user speech from SpokenTOD. Further training details are provided in Appendix B.2.

4.2 Baselines

We compare SpokenUS against human recordings and five end-to-end omni models at two

scales: Qwen2.5-Omni-3B (Xu et al., 2025a) and InteractiveOmni-4B (Tong et al., 2025), Qwen2.5-Omni-7B (Xu et al., 2025a), GLM-4-Voice-9B (Zeng et al., 2024), and Qwen3-Omni-30B-A3B (Xu et al., 2025b). In all evaluations, the dialogue assistant is fixed as GPT-4.1-mini (OpenAI, 2025).

4.3 Evaluation Metrics

Unless otherwise noted, all evaluations are conducted on 100 dialogues randomly sampled from the SpokenWOZ test set (held out from training), which provides the database states necessary for interactive user-assistant conversations. We limit the subset to 100 dialogues due to the substantial inference and human evaluation cost of end-to-end spoken dialogue evaluation.

Goal Coverage. We report *Goal Alignment (GA)*, the percentage of dialogues in which *all* required slot values are successfully conveyed, and *Slot Match Rate (SMR)*, the percentage of required slots conveyed across all dialogues as a finer-grained complement. Both metrics are evaluated by GPT-4.1-mini (OpenAI, 2025), which has been shown to achieve high agreement with humans on slot verification in TOD (Shim et al., 2026).

Human MOS. We recruit 10 English-proficient evaluators and randomly sample 20 dialogues from the evaluation set with 100% GA. Each dialogue is rated by at least 3 evaluators on three 5-point scales: *Naturalness (N)*, measuring voice quality; *Conversational Flow (C)*, assessing role consistency as a help-seeker and naturalness of turn-by-turn engagement; and *Spoken User Behavior (S)*, evaluating holistic authenticity of spoken behavior. Full details are provided in Appendix C.8.

Intelligibility. We measure speech intelligibility via WER, computed between each model’s text transcript and its corresponding synthesized speech using Whisper-large-v3 (Radford et al., 2023). This

captures how faithfully the intended text is realized in the generated speech, which is particularly important for omni models that explicitly generate text before synthesizing speech.

Speaker Similarity. We measure speaker consistency using cosine similarity between speaker embeddings extracted by wavlm-base-plus-sv (Chen et al., 2022). $SIM(s_0, s_i)$ measures similarity between the first turn and each subsequent turn, reflecting how consistently the simulator maintains speaker identity throughout the dialogue. $SIM(s_{i-1}, s_i)$ measures similarity between consecutive turns, capturing turn-level voice consistency.

5 Main Results

Table 3 presents results across goal coverage, human MOS, intelligibility and speaker similarity.

Goal Coverage. SpokenUS achieves a GA of 0.82, on par with Qwen2.5-Omni-7B (0.80) while substantially outperforming comparable-sized models (Qwen2.5-Omni-3B: 0.36; InteractiveOmni-4B: 0.59). Only Qwen3-Omni-30B-A3B surpasses it (0.93), at $10\times$ the model size and $\sim 20M$ hours of training data, demonstrating that TOD grounding enables strong goal coverage at small scale. Failure cases are analyzed in Appendix C.3.

Human MOS. SpokenUS achieves an average MOS of 4.06, substantially outperforming all baselines including Qwen3-Omni-30B-A3B (3.18) despite training on orders of magnitude less audio data ($\sim 1K$ vs. $\sim 20M$ hours). SpokenUS also scores higher than human recordings (3.67) across all three criteria; we note that SpokenWOZ recordings were collected at 8kHz under telephone-channel conditions, which may partially account for this gap (see Appendix C.9). These results demonstrate that SpokenUS generates speech that is perceived as natural and human-like (see Fig 21).

Intelligibility. SpokenUS achieves a WER of 11.36%, comparable to Qwen2.5-Omni-7B (10.53%) and substantially lower than the remaining baselines (up to 15.68%). Despite generating text before speech, omni models show higher WER, indicating that their synthesized speech often deviates from their own generated text.

Speaker Similarity. We interpret SIM relative to the natural range of variation observed in human speech, rather than as a monotonic quality measure. Human recordings exhibit relatively lower similarity scores (0.84 and 0.88), reflecting natural variation in emotion and prosody across turns. Omni

Label	Correct	Early	Confused	Missed
Turn-end	66.0%	10.4%	16.4%	7.2%
Barge-in	58.6%	18.0%	11.0%	12.4%

Table 4: Interruption performance on SpokenUS.

models tend to produce overly uniform speech, yielding higher scores. SpokenUS closely approximates the human level (0.93 and 0.92), indicating that its explicit modeling of spoken behaviors introduces natural speaker variation. While Qwen2.5-Omni-3B shows similarly lower SIM scores (0.90 and 0.92), its significantly degraded speech quality (MOS: 2.34) suggests that this score does not reflect natural speaker diversity.

6 Analysis

6.1 Interrupt Detection

To evaluate whether SpokenUS correctly identifies when to speak during streaming assistant speech, we randomly sample 500 assistant turns ending with *turn-end* and 500 ending with *barge-in* from the SpokenTOD test set, processing each utterance sequentially from the beginning. We categorize outcomes into four classes: **Correct** (correct label within the trigger window), **Early** (fired before the trigger window), **Confused** (wrong non-*listen* label), and **Missed** (*listen* throughout). Table 4 reports the turn-taking performance of SpokenUS. Turn-end detection achieves 66.0% accuracy with a missed rate of 7.2%, indicating that the model reliably recognizes when the assistant has finished speaking. Barge-in detection is more challenging at 58.6%, with a higher missed rate (12.4%), reflecting the inherent difficulty of proactive interruption where the model must decide to intervene based on partial context. Notably, a substantial portion of errors stem from confusion between *barge-in* and *turn-end*; when reduced to binary detection of whether to speak or continue listening (*listen*), effective accuracy rises to 82.4% for turn-end and 69.6% for barge-in, suggesting that the primary difficulty lies in disambiguating the two speaking behaviors rather than in detecting the moment to speak itself. Scoring and aggregation method details are in Appendix C.11.

6.2 Slot Disclosure Timing

Figure 3 shows cumulative slot coverage over user turns—the proportion of goal slots that have appeared in the dialogue by each turn, normalized by

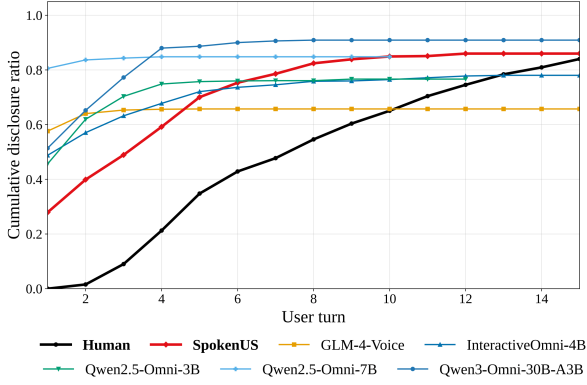


Figure 3: Cumulative goal slot coverage over user turns.

Model	Text	Cascaded
Qwen2.5-3B-Instruct	70.41	68.24 $\downarrow 2.17$
Qwen2.5-Omni-3B	83.63	72.16 $\downarrow 11.47$
SpokenUS	79.71	53.54 $\downarrow 26.17$

Table 5: Final Slot F1 scores on 100% GA dialogues.

the total number of slots per dialogue. We track only slot appearance rather than value correctness to isolate disclosure timing. Omni models tend to disclose slots within the first few turns and plateau early, whereas human speakers reveal them more gradually over the course of the dialogue. SpokenUS closely follows the human pattern, exhibiting a steady increase in slot coverage throughout the dialogue rather than front-loading information.

Beyond slot disclosure timing, this incremental disclosure also reduces generation latency. SpokenUS produces substantially shorter outputs per turn (35 vs. 88–202 chars/turn), yielding the lowest text-generation latency among all baselines (0.84s) and end-to-end latency competitive with comparable-sized models (Appendix C.7).

6.3 Impact on Agent Performance

We analyze how spoken user behaviors affect downstream agent performance. To construct a controlled evaluation set, we run each simulator 10 times and retain only dialogues where 100% GA is achieved across all runs, yielding 84 dialogues. We evaluate Final Turn Slot F1 against gold annotations from SpokenWOZ under two configurations: **Text**, where the agent receives the user transcript directly, and **Cascaded**, where user speech is first transcribed via ASR. As shown in Table 5, SpokenUS already poses a greater challenge at the transcript level (73.8 vs. 86.8), indicating that human-like content patterns such as cross-turn slots and gradual slot disclosure are inherently harder for the

agent to track. The gap widens substantially in the Cascaded setting (SpokenUS: −24.3 vs. Qwen2.5-Omni-3B: −3.7), indicating that spoken user behaviors themselves pose a significant challenge for the agents—not an artifact of speech quality, as evidenced by SpokenUS’s low WER and high Human MOS (§5). Notably, error analysis reveals that when ASR misrecognizes slot values—particularly those accompanied by disfluencies—agents tend to accept the erroneous transcription rather than proactively requesting confirmation from the user. Further details are provided in Appendix C.12.

7 Related Works

User Simulation for TOD. User simulation has long been a cornerstone of TOD research, enabling agents to be trained under diverse conditions without the cost of collecting real interactions (Eckert et al., 1997). Early simulators (Schatzmann et al., 2007; Chandramohan et al., 2012) modeled user intent and dialogue acts via hand-crafted rules or learned policies. Recent studies (Sekulic et al., 2024; Shim et al., 2026) have leveraged LLMs to generate more naturalistic user utterances, improving coverage and diversity. However, these simulators operate exclusively in text and do not account for the spontaneous spoken behaviors, such as disfluencies (Shriberg, 1994), barge-ins (Khouzaimi et al., 2015), and emotional prosody (Schröder, 2001), that emerge naturally in spoken interaction.

TOD Datasets. To support spoken TOD research, several spoken TOD datasets have been proposed. Early benchmarks such as DSTC2 (Henderson et al., 2014) and DSTC10 (Kim et al., 2021) evaluated TOD systems on spoken conversations but released only ASR transcripts and covered narrow domains. SpokenWOZ (Si et al., 2023) advanced the field with large-scale human-to-human recordings and fully annotated dialogue states. Nevertheless, human data collection remains costly and difficult to scale to new domains. Moreover, no existing work provides a systematic pipeline for augmenting text-based TOD corpora such as MultiWOZ (Budzianowski et al., 2018), SGD (Rastogi et al., 2020), ABCD (Chen et al., 2021), and Taskmaster (Byrne et al., 2019) with diverse spoken user behaviors across varied speakers and domains.

Spoken Dialogue Modeling. Early spoken dialogue systems relied on cascaded ASR–LLM–TTS pipelines (Huang et al., 2024). Such pipelines suf-

fer from compounding ASR errors (Sunder et al., 2022), substantial latency from sequential processing, and loss of non-linguistic cues such as emotional prosody and speaker characteristics (Défossez et al., 2024). Generative spoken dialogue models (Nguyen et al., 2023; Lee et al., 2025) offer an alternative by modeling full conversations directly in the speech domain, but are designed to synthesize entire dialogues rather than engage interactively, making them unsuitable for turn-by-turn user simulation. End-to-end omni models (Xu et al., 2025a,b; Zeng et al., 2024; Tong et al., 2025) and full-duplex models such as Moshi (Défossez et al., 2024), LSLM (Ma et al., 2025), and Person-aPlex (Roy et al., 2026) enable interactive spoken dialogue by processing and generating speech directly. However, all of these models are designed for system-side generation rather than user-side simulation, and none provide architectural support for proactive turn-taking behaviors.

8 Conclusion

We introduced SpokenTOD and SpokenUS to address the lack of realistic spoken user simulation for task-oriented dialogue. SpokenTOD provides a large-scale spoken TOD dataset enhanced with diverse user behaviors via an automated pipeline, and SpokenUS leverages it to train a spoken user simulator grounded in TOD knowledge with dedicated support for proactive turn-taking. Experimental results demonstrate that SpokenUS achieves strong goal coverage, while substantially outperforming all baselines in speech naturalness. Our agent robustness analysis further shows that SpokenUS’s spoken behaviors pose a substantially greater challenge to downstream agents than existing omni models, underscoring the importance of realistic spoken user simulation for building and evaluating robust task-oriented dialogue systems.

9 Limitations

SpokenTOD is constructed via an automated pipeline over existing text-based TOD corpora, which limits its domain coverage; however, the pipeline can be readily applied to other TOD datasets to extend coverage. Both SpokenTOD and SpokenUS are currently limited to English, though the pipeline can be extended to other languages as long as spoken user behavior annotations and a TOD dataset in the target language are available. Additionally, SpokenUS models a predefined set of spoken behaviors; phenomena such as laughter, overlapping speech, or code-switching are not yet covered. Moreover, SpokenTOD relies on single TTS-synthesized speech, which may not fully capture the acoustic variability of real-world environments; we mitigate this by incorporating SpokenWOZ as a source of authentic human speech, and note that our synthesis pipeline achieves a WER of 4.69% (§2.2), confirming that the generated speech is sufficiently intelligible for downstream training and evaluation. Furthermore, SpokenTOD does not model adverse acoustic conditions commonly encountered in real-world deployments, such as background noise, reverberation, and low-bandwidth channels. Since the synthesized audio already exists as standalone files, such conditions can be introduced by simply applying standard acoustic augmentation (e.g., noise injection, room impulse response convolution) over the existing SpokenTOD speech, requiring no modification to the synthesis pipeline. Likewise, the same augmentation can be directly applied to SpokenUS’s generated speech at inference time, enabling it to produce more realistic user speech without retraining. Finally, although SpokenUS achieves higher goal coverage than all comparable-sized baselines, its remaining failures concentrate in request omissions after domain transitions, pointing to multi-domain tracking as a targeted direction for future improvement—per-domain completion verification during training, or reinforcement learning with rewards that penalize premature transitions.

References

- Amazon. 2026. Alexa. <https://alexa.amazon.com/about>. Accessed 2026-03-04.
- Apple. 2024. Apple intelligence: Ai for the rest of us. <https://www.apple.com/apple-intelligence/>. Accessed 2026-03-04.

674	Jennifer E. Arnold. 2013. Information status relates to production, distribution, and comprehension . <i>Frontiers in Psychology</i> , Volume 4 - 2013.	730
675		731
676		732
677	Evelina Bakhturina, Yang Zhang, and Boris Ginsburg. 2022. Shallow Fusion of Weighted Finite-State Transducer and Language Model for Text Normalization . In <i>Interspeech 2022</i> , pages 491–495.	733
678		734
679		735
680		736
681	Paweł Budzianowski, Tsung-Hsien Wen, Bo-Hsiang Tseng, Iñigo Casanueva, Stefan Ultes, Osman Ramadan, and Milica Gašić. 2018. MultiWOZ - a large-scale multi-domain Wizard-of-Oz dataset for task-oriented dialogue modelling . In <i>Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing</i> , pages 5016–5026, Brussels, Belgium. Association for Computational Linguistics.	737
682		738
683		739
684		740
685		741
686		742
687		743
688		744
689	Bill Byrne, Karthik Krishnamoorthi, Chinnadhurai Sankar, Arvind Neelakantan, Ben Goodrich, Daniel Duckworth, Semih Yavuz, Amit Dubey, Kyu-Young Kim, and Andy Cedilnik. 2019. Taskmaster-1: Toward a realistic and diverse dialog dataset . In <i>Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)</i> , pages 4516–4525, Hong Kong, China. Association for Computational Linguistics.	745
690		746
691		747
692		748
693		749
694		750
695		751
696		752
697		753
698		754
699		755
700	Senthilkumar Chandramohan, Matthieu Geist, Fabrice Lefevre, and Olivier Pietquin. 2012. Behavior specific user simulation in spoken dialogue systems. In <i>Speech Communication; 10. ITG Symposium</i> , pages 1–4.	756
701		757
702		758
703		759
704		760
705	Derek Chen, Howard Chen, Yi Yang, Alexander Lin, and Zhou Yu. 2021. Action-based conversations dataset: A corpus for building more in-depth task-oriented dialogue systems . In <i>Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies</i> , pages 3002–3017, Online. Association for Computational Linguistics.	761
706		762
707		763
708		764
709		765
710		766
711		767
712		768
713	Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Yu, and Furu Wei. 2022. Wavlm: Large-scale self-supervised pre-training for full stack speech processing . <i>IEEE Journal of Selected Topics in Signal Processing</i> , 16(6):1505–1518.	769
714		770
715		771
716		772
717		773
718		774
719		775
720		776
721	Herbert H Clark. 1996. <i>Using language</i> . Cambridge university press.	777
722		778
723	Herbert H Clark and Susan E Brennan. 1991. Grounding in communication.	779
724		780
725	Alexandru Coca, Bo-Hsiang Tseng, Peter Boothroyd, Jianpeng Cheng, Zhenxing Zhang, Mark Gaynor, Joe Stacey, Tristan Guigue, Héctor Martínez Alonso, Diarmuid Ó Séaghdha, and 1 others. 2025. Pytod: Programmable task-oriented dialogue with execution	781
726		782
727		783
728		784
729		785
		786
	feedback. In <i>Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue</i> , pages 93–112.	730
		731
		732
	Martin Corley, Lucy J. MacGregor, and David I. Donaldson. 2007. It’s the way that you, er, say it: Hesitations in speech affect language comprehension . <i>Cognition</i> , 105(3):658–668.	733
		734
		735
		736
	Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. 2024. Moshi: a speech-text foundation model for real-time dialogue. <i>arXiv preprint arXiv:2410.00037</i> .	737
		738
		739
		740
		741
	Zhihao Du, Changfeng Gao, Yuxuan Wang, Fan Yu, Tianyu Zhao, Hao Wang, Xiang Lv, Hui Wang, Chongjia Ni, Xian Shi, and 1 others. 2025. Cosyvoice 3: Towards in-the-wild speech generation via scaling-up and post-training. <i>arXiv preprint arXiv:2505.17589</i> .	742
		743
		744
		745
		746
		747
	W. Eckert, E. Levin, and R. Pieraccini. 1997. User modeling for spoken dialogue system evaluation . In <i>1997 IEEE Workshop on Automatic Speech Recognition and Understanding Proceedings</i> , pages 80–87.	748
		749
		750
		751
	Shutong Feng, Nurul Lubis, Christian Geisshauser, Hsien-chin Lin, Michael Heck, Carel van Niekerk, and Milica Gasic. 2022. EmoWOZ: A large-scale corpus and labelling scheme for emotion recognition in task-oriented dialogue systems . In <i>Proceedings of the Thirteenth Language Resources and Evaluation Conference</i> , pages 4096–4113, Marseille, France. European Language Resources Association.	752
		753
		754
		755
		756
		757
		758
		759
	Elizabeth Han, Dezhi Yin, and Han Zhang. 2023. Bots with feelings: should ai agents express positive emotion in customer service? <i>Information Systems Research</i> , 34(3):1296–1311.	760
		761
		762
		763
	Matthew Henderson, Blaise Thomson, and Jason D. Williams. 2014. The second dialog state tracking challenge . In <i>Proceedings of the 15th Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL)</i> , pages 263–272, Philadelphia, PA, U.S.A. Association for Computational Linguistics.	764
		765
		766
		767
		768
		769
		770
	Hangrui Hu, Xinfu Zhu, Ting He, Dake Guo, Bin Zhang, Xiong Wang, Zhifang Guo, Ziyue Jiang, Hongkun Hao, Zishan Guo, and 1 others. 2026. Qwen3-tts technical report. <i>arXiv preprint arXiv:2601.15621</i> .	771
		772
		773
		774
	Rongjie Huang, Mingze Li, Dongchao Yang, Jia-tong Shi, Xuankai Chang, Zhenhui Ye, Yuning Wu, Zhiqing Hong, Jiawei Huang, Jinglin Liu, Yi Ren, Yuexian Zou, Zhou Zhao, and Shinji Watanabe. 2024. Audiogpt: understanding and generating speech, music, sound, and talking head . In <i>Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence, AAAI’24/IAAI’24/EAAI’24</i> . AAAI Press.	775
		776
		777
		778
		779
		780
		781
		782
		783
		784
		785
		786

- Hatim Khouzaimi, Romain Laroche, and Fabrice Lefèvre. 2015. [Turn-taking phenomena in incremental dialogue systems](#). In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 1890–1895, Lisbon, Portugal. Association for Computational Linguistics.
- Seokhwan Kim, Yang Liu, Di Jin, Alexandros Papanagelis, Karthik Gopalakrishnan, Behnam Hedayatnia, and Dilek Hakkani-Tür. 2021. [“how robust r u?”: Evaluating task-oriented dialogue systems on spoken conversations](#). In *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 1147–1154.
- Jungil Kong, Jaehyeon Kim, and Jaekyoung Bae. 2020. [Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 17022–17033. Curran Associates, Inc.
- Jacqueline C. Kowtko and Patti J. Price. 1989. [Data collection and analysis in the air travel planning domain](#). In *Speech and Natural Language: Proceedings of a Workshop Held at Cape Cod, Massachusetts, October 15-18, 1989*.
- Sehun Lee, Kang-wook Kim, and Gunhee Kim. 2025. [Behavior-SD: Behaviorally aware spoken dialogue generation with large language models](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 9574–9593, Albuquerque, New Mexico. Association for Computational Linguistics.
- Hongcheng Liu, Yixuan Hou, Heyang Liu, Yuhao Wang, Yanfeng Wang, and Yu Wang. 2025. [Vocalbench-df: A benchmark for evaluating speech llm robustness to disfluency](#). *Preprint*, arXiv:2510.15406.
- Ziyang Ma, Yakun Song, Chenpeng Du, Jian Cong, Zhuo Chen, Yuping Wang, Yuxuan Wang, and Xie Chen. 2025. [Language model can listen while speaking](#). *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(23):24831–24839.
- Jarana Manotumruksa, Jeff Dalton, Edgar Meij, and Emine Yilmaz. 2021. [Improving dialogue state tracking with turn-based loss function and sequential data augmentation](#). In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 1674–1683, Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Benjamin Marie. 2023. [Disfluency generation for more robust dialogue systems](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 11479–11488, Toronto, Canada. Association for Computational Linguistics.
- Fergus McInnes and David Attwater. 2004. [Turn-taking and grounding in spoken telephone number transfers](#). *Speech Communication*, 43(3):205–223.
- Fabian Mentzer, David Minnen, Eiríkur Agustsson, and Michael Tschannen. 2024. [Finite scalar quantization: VQ-VAE made simple](#). In *The Twelfth International Conference on Learning Representations*.
- Marie Meteer and 1 others. 1995. Dysfluency annotation stylebook for the savitchboard corpus. Technical report, Technical report.
- Tu Anh Nguyen, Eugene Kharitonov, Jade Copet, Yossi Adi, Wei-Ning Hsu, Ali Elkahky, Paden Tomasello, Robin Algayres, Benoît Sagot, Abdelrahman Mohamed, and Emmanuel Dupoux. 2023. [Generative spoken dialogue language modeling](#). *Transactions of the Association for Computational Linguistics*, 11:250–266.
- Vahid Noroozi, Somshubra Majumdar, Ankur Kumar, Jagadeesh Balam, and Boris Ginsburg. 2024. [Stateful conformer with cache-based inference for streaming automatic speech recognition](#). In *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 12041–12045.
- OpenAI. 2025. Introducing gpt-4.1 in the api. <https://openai.com/index/gpt-4-1/>. Accessed 2026-03-04.
- Andrew Ortony, Gerald L Clore, and Allan Collins. 2022. *The cognitive structure of emotions*. Cambridge university press.
- Tatiana Passali, Thanassis Mavropoulos, Grigorios Tsoumakas, Georgios Meditskos, and Stefanos Vrochidis. 2022. [LARD: Large-scale artificial disfluency generation](#). In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 2327–2336, Marseille, France. European Language Resources Association.
- Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, and 25 others. 2025. [Qwen2.5 technical report](#). *Preprint*, arXiv:2412.15115.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine Mcleavey, and Ilya Sutskever. 2023. [Robust speech recognition via large-scale weak supervision](#). In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 28492–28518. PMLR.
- Abhinav Rastogi, Xiaoxue Zang, Srinivas Sunkara, Raghav Gupta, and Pranav Khaitan. 2020. [Towards scalable multi-domain conversational agents: The schema-guided dialogue dataset](#). *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05):8689–8696.
- Rajarshi Roy, Jonathan Raiman, Sang-gil Lee, Teodor-Dumitru Ene, Robert Kirby, Sungwon Kim, Jaehyeon

900	Kim, and Bryan Catanzaro. 2026. Personaplex:	953
901	Voice and role control for full duplex conversational	954
902	speech models. <i>arXiv preprint arXiv:2602.06053</i> .	955
903	Harvey Sacks, Emanuel A Schegloff, and Gail Jefferson.	956
904	1974. A simplest systematics for the organization of	957
905	turn-taking for conversation. <i>language</i> , 50(4):696–	958
906	735.	959
907	Jost Schatzmann, Blaise Thomson, Karl Weilhammer,	960
908	Hui Ye, and Steve Young. 2007. Agenda-based user	961
909	simulation for bootstrapping a POMDP dialogue sys-	962
910	tem. In <i>Human Language Technologies 2007: The</i>	963
911	<i>Conference of the North American Chapter of the</i>	964
912	<i>Association for Computational Linguistics; Compan-</i>	965
913	<i>ion Volume, Short Papers</i> , pages 149–152, Rochester,	966
914	New York. Association for Computational Linguis-	
915	tics.	
916	Marc Schröder. 2001. Emotional speech synthesis: a	
917	review. In <i>Interspeech</i> , volume 2001, pages 561–564.	
918	Ivan Sekulic, Silvia Terragni, Victor Guimarães, Nghia	
919	Khau, Bruna Guedes, Modestas Filipavicius, Andre	
920	Ferreira Manso, and Roland Mathis. 2024. Re-	
921	liable LLM-based user simulator for task-oriented	
922	dialogue systems. In <i>Proceedings of the 1st Work-</i>	
923	<i>shop on Simulating Conversational Intelligence in</i>	
924	<i>Chat (SCI-CHAT 2024)</i> , pages 19–35, St. Julians,	
925	Malta. Association for Computational Linguistics.	
926	Priyanka Sen. 2020. Speech disfluencies occur at higher	
927	perplexities. In <i>Proceedings of the Workshop on</i>	
928	<i>the Cognitive Aspects of the Lexicon</i> , pages 92–97,	
929	Online. Association for Computational Linguistics.	
930	Gyuhyeon Seo, Jungwoo Yang, Junseong Pyo, Nalim	
931	Kim, Jonggeun Lee, and Yohan Jo. 2026. Simuhome:	
932	A temporal- and environment-aware benchmark for	
933	smart home LLM agents. In <i>The Fourteenth Interna-</i>	
934	<i>tional Conference on Learning Representations</i> .	
935	Jeonghoon Shim, Woojung Song, Cheyon Jin, Seung-	
936	won KooK, and Yohan Jo. 2026. Non-collaborative	
937	user simulators for tool agents. In <i>The Fourteenth In-</i>	
938	<i>ternational Conference on Learning Representations</i> .	
939	Elizabeth Shriberg. 1996. Disfluencies in switchboard.	
940	In <i>Proceedings of international conference on spoken</i>	
941	<i>language processing</i> , volume 96, pages 11–14. IEEE	
942	Philadelphia, PA.	
943	Elizabeth Ellen Shriberg. 1994. Preliminaries to a the-	
944	ory of speech disfluencies. <i>Doctoral dissertation,</i>	
945	<i>University of California at Berkeley</i> .	
946	Shuzheng Si, Wentao Ma, Haoyu Gao, Yuchuan Wu,	
947	Ting-En Lin, Yinpei Dai, Hangyu Li, Rui Yan, Fei	
948	Huang, and Yongbin Li. 2023. Spokenwoz: A large-	
949	scale speech-text benchmark for spoken task-oriented	
950	dialogue agents. In <i>Advances in Neural Information</i>	
951	<i>Processing Systems</i> , volume 36, pages 39088–39118.	
952	Curran Associates, Inc.	
	Tanya Stivers, N. J. Enfield, Penelope Brown, Christina	
	Englert, Makoto Hayashi, Trine Heinemann, Gertie	
	Hoymann, Federico Rossano, Jan Peter de Ruiter,	
	Kyung-Eun Yoon, and Stephen C. Levinson. 2009.	
	Universals and cultural variation in turn-taking in	
	conversation. <i>Proceedings of the National Academy</i>	
	<i>of Sciences</i> , 106(26):10587–10592.	
	Vishal Sunder, Samuel Thomas, Hong-Kwang J. Kuo,	
	Jatin Ganhotra, Brian Kingsbury, and Eric Fosler-	
	Lussier. 2022. Towards end-to-end integration of	
	dialog history for improved spoken language under-	
	standing. In <i>ICASSP 2022 - 2022 IEEE International</i>	
	<i>Conference on Acoustics, Speech and Signal Process-</i>	
	<i>ing (ICASSP)</i> , pages 7497–7501.	
	Wenwen Tong, Hewei Guo, Dongchuan Ran, Jiang-	
	nan Chen, Jiefan Lu, Kaibin Wang, Keqiang Li, Xi-	
	aoux Zhu, Jiakui Li, Kehan Li, and 1 others. 2025.	
	Interactiveomni: A unified omni-modal model for	
	audio-visual multi-turn dialogue. <i>arXiv preprint</i>	
	<i>arXiv:2510.13747</i> .	
	Steven H Weinberger and Stephen A Kunath. 2011. The	
	speech accent archive: towards a typology of english	
	accents. <i>Language & Computers</i> , 73(1).	
	Silke Witt. 2015. Modeling user response timings	
	in spoken dialog systems. <i>Int. J. Speech Technol.</i> ,	
	18(2):231–243.	
	Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting	
	He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan,	
	Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and	
	Junyang Lin. 2025a. Qwen2.5-omni technical report.	
	<i>Preprint</i> , arXiv:2503.20215.	
	Jin Xu, Zhifang Guo, Hangrui Hu, Yunfei Chu, Xiong	
	Wang, Jinzheng He, Yuxuan Wang, Xian Shi, Ting	
	He, Xinfu Zhu, Yuanjun Lv, Yongqi Wang, Dake	
	Guo, He Wang, Linhan Ma, Pei Zhang, Xinyu	
	Zhang, Hongkun Hao, Zishan Guo, and 19 others.	
	2025b. Qwen3-omni technical report. <i>Preprint</i> ,	
	arXiv:2509.17765.	
	An Yang, Anfeng Li, Baosong Yang, Beichen Zhang,	
	Binyuan Hui, Bo Zheng, Bowen Yu, Chang	
	Gao, Chengen Huang, Chenxu Lv, and 1 others.	
	2025. Qwen3 technical report. <i>arXiv preprint</i>	
	<i>arXiv:2505.09388</i> .	
	Jingfeng Yang, Diyi Yang, and Zhaoran Ma. 2020. Plan-	
	ning and generating natural and diverse disfluent texts	
	as augmentation for disfluency detection. In <i>Proceed-</i>	
	<i>ings of the 2020 Conference on Empirical Methods</i>	
	<i>in Natural Language Processing (EMNLP)</i> , pages	
	1450–1460, Online. Association for Computational	
	Linguistics.	
	Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong	
	Wang, Shengmin Jiang, Lei Zhao, Yuxiao Dong, and	
	Jie Tang. 2024. Glm-4-voice: Towards intelligent	
	and human-like end-to-end spoken chatbot. <i>arXiv</i>	
	<i>preprint arXiv:2412.02612</i> .	

Sai Zhang, Yuwei Hu, Yuchuan Wu, Jiaman Wu, Yongbin Li, Jian Sun, Caixia Yuan, and Xiaojie Wang. 2022. [A slot is not built in one utterance: Spoken language dialogs with sub-slots](#). In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 309–321, Dublin, Ireland. Association for Computational Linguistics.

Yang Zhang, Evelina Bakhturina, and Boris Ginsburg. 2021. NeMo (Inverse) Text Normalization: From Development to Production. In *Interspeech 2021*, pages 4857–4859.

A SpokenTOD Details

In this section, we provide a detailed breakdown of the data processing pipeline used to construct SpokenTOD.

A.1 Dataset Details

This section provides details about the five datasets used to construct SpokenTOD.

SGD. Schema-Guided Dialogue (SGD) dataset (Rastogi et al., 2020) contains over 20K task-oriented conversations spanning 20 domains such as travel, media, banking, and events. SGD adopts a schema-guided mechanism in which each service is described via natural language definitions of its intents and slots, rather than a fixed ontology. Dialogues were generated via a rule-based simulator and subsequently paraphrased by crowd workers to improve naturalness, with slot annotations recovered automatically via string matching.

Taskmaster-2. Taskmaster-2 (Byrne et al., 2019) consists of 17K spoken two-person dialogues collected via a Wizard-of-Oz (WOZ) methodology across seven domains: restaurants, food ordering, movies, hotels, flights, music, and sports. User utterances were transcribed from audio recordings of crowdsourced workers via ASR. However, the original audio recordings are not publicly released.

ABCD Action-Based Conversations Dataset (ABCD) (Chen et al., 2021) is a fully-labeled human-to-human dataset containing over 10K customer service dialogues with 55 distinct user intents. Data was collected via an Expert Live Chat system pairing trained call center agents with crowdsourced users, with action labels annotated by agents in real time during the conversation.

EmoWOZ. EmoWOZ (Feng et al., 2022) is a large-scale emotion-annotated corpus built on MultiWOZ, containing over 11K dialogues with 83K user utterance-level emotion labels. Each utterance was annotated by three crowd workers via majority voting. The labelling scheme is grounded in the Ortony–Clore–Collins (OCC) model (Ortony et al., 2022), defining seven emotion categories tailored to task-oriented settings: NEUTRAL, FEARFUL, DISSATISFIED, APOLOGETIC, ABUSIVE, EXCITED, and SATISFIED. We adopt this label set for emotion annotation across all source datasets in SpokenTOD, using EmoWOZ’s human-annotated labels directly where available.

SpokenWOZ. SpokenWOZ (Si et al., 2023) is the first large-scale speech-text benchmark for spoken TOD, comprising 5.7K dialogues, 203K turns, and 249 hours of audio from human-to-human spoken conversations across 8 domains. It introduces spoken-specific challenges such as cross-turn slot filling and word-by-word reasoning, which are absent in text-only datasets. As the only natively spoken source dataset in SpokenTOD, SpokenWOZ provides real acoustic and prosodic grounding for our dataset.

Dataset Licenses All datasets are used in accordance with their respective licenses: ABCD and Taskmaster-2 under CC BY 4.0, SpokenWOZ and EmoWOZ under CC BY-NC 4.0, SGD under CC BY-SA 4.0, and the Speech Accent Archive under CC BY-NC-SA 2.0. These datasets were originally released for academic research in task-oriented dialogue systems, and our use for benchmark evaluation is consistent with this intended purpose. Our dataset and model are provided for research purposes only.

A.2 Dataset Unification

To construct SpokenTOD from source datasets with varying formats, we unify each dataset into a common schema with two components: a user goal $G = (T, S)$, and slot value positions within each utterance for injecting disfluency near slot values (Section 2.1). G consists of a natural-language goal text T and a structured goal S ; S records for each sub-goal, the domain, intent type, and slot-value pairs (e.g., `find_and_book` in the `hotel` domain with `area: south`, `pricerange: moderate`). As each source dataset has its own annotation format and characteristics, we apply dataset-specific preprocessing procedures as follows.

SGD. SGD annotates each turn with start and end positions for slot values, which we use directly. Since SGD does not provide explicit user goals G , we reconstruct it from turn-level annotations accumulated across the dialogue: S is constructed by mapping turn-level intent and slot-value pairs into the unified schema, from which T is generated by filling a rule-based template with the collected intents and constraints.

TM-2. TM-2 annotates each turn with `start_index` and `end_index` fields marking slot value positions, which we use directly. Since

TM-2 does not provide explicit user goals G or intents, we reconstruct it from slot-value arguments collected across turns: S is constructed by normalizing these into the unified schema, from which T is generated by filling a rule-based template with the collected arguments.

ABCD. ABCD pairs each dialogue with a delexicalized version in which slot values are replaced with typed placeholders (e.g., `<username>`, `<email>`, `<order_id>`). We recover slot value positions by aligning the delexicalized and original utterances via prefix/suffix matching; for values not covered by placeholders, we match them directly from the metadata. Since ABCD does not provide explicit user goals G , we reconstruct it from the scenario metadata: S is constructed by mapping domain and intent labels with recovered slot-value pairs into the unified schema, from which T is generated by filling a rule-based template with the customer prompt and scenario constraints.

EmoWOZ. EmoWOZ extends MultiWOZ with per-turn emotion labels while inheriting its goal and slot annotations. We use the slot value positions from MultiWOZ directly. G is taken directly from the MultiWOZ goal annotations: S is mapped into the unified schema, and T is the natural-language goal description provided by MultiWOZ. Emotion labels are used as provided.

SpokenWOZ. SpokenWOZ provides dialogue state and dialogue act annotations from which we derive slot value positions. G is constructed following the same procedure as EmoWOZ, using SpokenWOZ’s goal annotations recorded in the MultiWOZ format.

A.3 Cross-Turn Slots Details

Cross-turn slots represent slot values that span multiple turns rather than in a single utterance, common with long alphanumeric sequences (phone numbers, email addresses, reservation IDs). We apply type-specific segmentation to all segmentable slot values: numeric sequences are split into 3–4 digit chunks, email addresses are decomposed into components with explicit punctuation vocalization (e.g., "dot", "at"), and alphanumeric codes separate letters from digits. To model realistic error patterns, we inject mistakes with $P_{\text{error}} = 0.20$, following the 20% error rate used in SpokenWOZ. This generates correction sub-dialogues where the user dictates an incorrect segment at turn t , then

self-corrects at turn $t+1$ (e.g., "Wait, I meant..."), and the assistant acknowledges the correction.

A.4 Barge-in Details

Barge-in refers to user interruptions during system speech, occurring in up to 25% of responses in spoken dialogue systems (Witt, 2015). Following the turn-taking taxonomy of Khouzaimi et al. (2015), we categorize barge-in by the assistant’s utterance state and user’s response type: (1) **Error recovery** (G_INCOHERENT): the user interrupts to correct inaccurate information in the assistant’s utterance (e.g., "No, I said Paris, not London"); (2) **Clarification** (G_FAIL): the user interrupts due to comprehension failure, expressing general confusion or requesting specific details; (3) **Efficiency** (G_SUFFICIENT): the user interrupts once enough information has been conveyed to move the conversation forward (e.g., "Yes, let’s book it").

We apply barge-in augmentation to 25% of randomly sampled user turns via a four-stage protocol. We first sample turns and uniformly assign one of the three barge-in types, along with one of three response styles (Khouzaimi et al., 2015): **Implicit** (REF_IMPL), where the user responds with minimal acknowledgment (e.g., "Aha"); **Raw** (REF_RAW), where the user directly references a specific element (e.g., "Ok, Sunday"); and **Interpreted** (REF_INTERP), where the user adds their own interpretation (e.g., "Yeah, Sunday works – that’s my only free day"). We then prompt Qwen3-32B (Yang et al., 2025) to assess whether the assistant’s utterance is a suitable target for the selected barge-in type, i.e., whether it contains incoherent information, unclear content, or sufficient information to proceed. If valid, the model generates a modified user turn that naturalistically implements the selected barge-in behavior and response style with appropriate discourse markers (e.g., "Wait", "Actually"). Detailed prompts for each combination of barge-in type and response style are provided in Figures 4–5 (Error Recovery), 6–7 (Clarification), and 8–10 (Efficiency).

A.5 Disfluency Details

Scope and sampling. We follow the exponential length-dependent model from Shriberg (1996), where the probability of an utterance being disfluent increases with its length L (measured in words):

$$P(\text{disfluent} \mid L) = 1 - b^L \quad (2)$$

where b represents the word-level fluency rate. Following Shriberg (1996), we set $b = 0.9453$ based on the AMEX corpus (Kowtko and Price, 1989), as it represents customer service interactions similar to task-oriented dialogue. This formulation ensures that longer utterances are more likely to contain disfluencies, reflecting patterns observed in spontaneous dialogue where speakers are more prone to self-corrections and hesitations in extended turns.

When an utterance is sampled to be disfluent, we uniformly sample one type from the six disfluency categories defined in Table 1: [FP], [DM], [EDIT], [REP], [COR], and [RST]. Following prior findings that disfluencies increase under higher planning and lexical retrieval difficulty (Corley et al., 2007; Arnold, 2013; Sen, 2020), we inject them more frequently near slot values, as retrieving and articulating domain-specific entities demands greater cognitive effort. For all disfluency types except corrections, we insert them with 50% probability near slots (within a 2-word window) and 50% probability uniformly across the utterance, while corrections [COR] exclusively target slot values.

Disfluency placement. We follow the annotation guidelines from the Switchboard Corpus (Meteer et al., 1995) for linguistically motivated disfluency placement. Filled pauses [FP], discourse markers [DM], and editing terms [EDIT] are inserted before the target word. Repetitions [REP] duplicate the target span. Corrections [COR] replace the original slot value with an alternative. Restarts [RST] abandon the current utterance structure and begin a rephrased continuation.

A.6 Emotion Annotation Details

To support emotion-conditioned speech synthesis, we annotate each user turn with seven emotion labels: Neutral, Fearful, Dissatisfied, Apologetic, Abusive, Excited, and Satisfied. We use Qwen3-32B (Yang et al., 2025) as our primary annotator. Annotation is skipped for EmoWOZ, as it already provides emotion labels. For turns that are cross-turn segments, we skip annotation and instead inherit the emotion label from the most recent non-segment user turn.

To ensure the reliability of LLM-based annotation, we validated Qwen3-32B against the ground truth labels of EmoWOZ. We sampled 50 instances per category (with 19 instances for Abusive due to its limited occurrence in the dataset, totaling 319 samples) and compared the model’s predictions

with the original labels. Qwen3-32B achieved an overall accuracy of 0.75, demonstrating its capability to capture affective states in task-oriented interactions.

A.7 Speech Synthesis Details

Text Normalization. Task-oriented dialogues frequently contain numbers, dates, times, addresses, and other entities that require normalization before speech synthesis. For instance, "*I need a table for 2 at 7:30pm*" must be converted to "*I need a table for two at seven thirty pm*" for proper pronunciation. We apply NeMo-text-processing (Zhang et al., 2021; Bakhturina et al., 2022) to all utterances before synthesis. Key normalization patterns include:

- **Numbers:** Cardinal (e.g., "123" → "one hundred twenty three") and ordinal forms (e.g., "1st" → "first")
- **Dates and times:** "01/15/2024" → "January fifteenth twenty twenty four", "7:30pm" → "seven thirty p m"
- **Currency and measurements:** "\$50" → "fifty dollars", "5km" → "five kilometers"
- **Abbreviations and acronyms:** "Dr." → "doctor", "NYC" → "N Y C" or "New York City" based on context

A.8 Emotion Conditioning

Qwen3-TTS conditions on emotion via instruction prompts of the form: "Please speak in a {emotion} tone."

We construct our emotion keyword set from the top-100 speaking styles in CosyVoice3’s pre-training data (Du et al., 2025). We first group these styles into seven coarse categories based on semantic similarity (neutral, fearful, dissatisfied, apologetic, abusive, excited, satisfied). We then evaluate the acoustic realization of each keyword through synthesis and select those that most effectively convey the intended emotion category, resulting in 3-6 keywords per category (Table 6). During synthesis, we randomly sample one keyword from the turn’s assigned category to introduce prosodic variation while maintaining category consistency.

A.9 Speaker Diversity

Reference audio clips from the Speech Accent Archive (Weinberger and Kunath, 2011) provide

Emotion Category	Style Keywords
NEUTRAL	calm, indifferent, patient, relaxed
FEARFUL	fearful, shocked, surprised
DISSATISFIED	angry, contempt, disgusted, defiant
APOLOGETIC	compassionate, selfless, humble
ABUSIVE	commanding, authoritative, merciless, loud, vengeful
EXCITED	adventurous, energetic, passionate, curious, creative, joyful
SATISFIED	proud, hopeful, happy, cheerful

Table 6: Emotion-to-keyword mapping for Qwen3-TTS instruction prompts. Keywords are drawn from the top-100 speaking styles in CosyVoice3’s pre-training data.

speaker identity and accent for zero-shot voice cloning. We filter candidate speakers to those with reference audio clips of at most 25 seconds, then organize them into four broad accent pools: Native, African, Indian, and Asian. A reference speaker is sampled via a three-stage procedure: (1) sample an accent pool using mixture weights derived from U.S. Census (2024) population estimates² (Table 7), (2) sample a country uniformly within the selected pool, and (3) randomly sample a speaker from that country. For demographic balancing, we stratify speaker age into four bins (10s, 20–30s, 40–50s, 60+) and sample each bin with equal probability (25%). Gender balance is also maintained equally.

The resulting pool comprises **542** unique speakers: 433 Native (79.89%), 57 Asian (10.52%), 38 Indian (7.01%), and 14 African (2.58%), reflecting the U.S. Census population weights used in sampling. For assistant utterances, we fix a pool of 10 Native-accent speakers (5 male / 5 female) and exclude them from the user pool to avoid identity overlap.

A.10 Dataset Statistics and Validation Details

A.10.1 Data Splits

We split SpokenTOD into train (39,430; 75%), validation (5,390; 10%), and test (7,570; 15%) sets.

A.10.2 Duration Verification

We verify that audio files exist for all dialogue turns in the metadata and that durations fall within expected ranges (0.3-30 seconds per turn). Total dataset duration is computed by summing individual file durations, validated through our pipeline which processes 1,208,554 audio files totaling 1,034 hours of speech.

²<https://www.census.gov/data/datasets/time-series/demo/popest/2020s-national-detail.html>

Pool	Countries	Population	Weight	# Spk.
Native	USA, UK, Canada, Australia, Ireland	197,875,652	74.57%	433
African	Ethiopia, Nigeria, Ghana, Senegal, Morocco	42,951,595	16.19%	14
Indian	India, Pakistan, Afghanistan, Bangladesh, Nepal	2,442,428	0.92%	38
Asian	China, South Korea, Japan, Philippines, Vietnam	22,080,844	8.32%	57
Total		265,350,519	100.00%	542

Table 7: Accent pool sampling weights based on U.S. Census (2024) estimates and resulting speaker counts. Speaker counts reflect the availability of qualifying recordings in the Speech Accent Archive after applying a 25-second duration filter.

A.10.3 Synthesized Audio Intelligibility

To validate the intelligibility of our synthesized speech, we measure WER on 500 randomly sampled dialogues using Whisper-large-v3 (Radford et al., 2023), covering all four accent groups. As shown in Table 8, our dataset achieves an overall WER of 4.69%, confirming that the synthesized audio is sufficiently intelligible.

Accent Group	WER (%)	# Utterances
African	5.08	1,340
Asian	3.77	1,242
Indian	4.95	1,355
Native	4.91	1,311
Overall	4.69	5,248

Table 8: ASR performance on synthesized speech using Whisper-large-v3

A.10.4 LLM Barge-in Appropriateness

To verify whether the LLM places barge-ins at contextually appropriate moments during augmentation (§A.4), we sample 20 barge-in turns from SpokenTOD and ask three annotators to judge whether each barge-in is contextually appropriate. A barge-in is judged appropriate when the assistant’s preceding utterance contains a clear trigger for interruption, such as information conflicting with the user’s stated goal (e.g., recommending a south-area restaurant when the user requested the centre). Across 60 total ratings (20 turns $\times$ 3 annotators),

83% of LLM-placed barge-ins are rated as contextually appropriate, indicating that the LLM understands the contextual conditions for barge-in: it constructs assistant turns whose preceding content naturally motivates user interruption. The audio-side realization of these barge-ins is further evaluated as part of the synthesized audio quality evaluation (Appendix C.10, Barge-in sub-dimension).

B SpokenUS Details

B.1 Operating Modes Details

Listening Mode. Each assistant turn begins with the delimiter `<|im_start|>assistant<BOA>`. SpokenUS processes each incoming speech token to obtain its last hidden state, which is passed to the turn-taking head. The head applies a softmax over three classes (listen, barge-in, and turn-end), and the resulting probabilities are aggregated via a linear-recency weighting scheme to determine whether to interrupt or yield.

Pre-scripting Mode. Once a barge-in or turn-end is detected, a turn delimiter (`<|im_end|><|im_start|>user<BOT>`) is inserted at that position. The last hidden state is then passed to the text head, which autoregressively generates the structured transcript until `<EOT>` is predicted.

Speaking Mode. Upon predicting `<EOT>`, `<BOA>` is appended and the last hidden state of each subsequent token is routed to the speech head. Speech token generation continues until `<EOA>` is predicted, at which point the generated discrete speech tokens are converted to audio via the two-stage synthesis pipeline described in Section 3.2.

B.2 SpokenUS Training Details

SpokenUS. SpokenUS is initialized from Qwen2.5-3B (Qwen et al., 2025) and fine-tuned in two stages. We augment the vocabulary with disfluency and emotion special tokens, as listed in Table 9. In Stage 1, the model is trained on full text-only SpokenTOD dialogues (both user and assistant turns) for 3 epochs, enabling the model to acquire task-oriented dialogue knowledge such as goal-directed response generation. In Stage 2, all three heads are jointly trained for 12k steps using the multi-task objective in Section 3.3, supervised on user turns only. Hyperparameters are summarized in Table 10.

CFM and HiFi-GAN. The CFM and HiFi-GAN are both initialized from CosyVoice3 (Du et al.,

Category	Tokens
Disfluency	[COR], [DM], [EDIT], [FP], [REP], [RST]
Emotion	< abusive >, < apologetic >, < excited >, < satisfied >, < fearful >, < neutral >, < dissatisfied >

Table 9: Disfluency and emotion related tokens added to the vocabulary.

Hyperparameter	Value
Backbone	Qwen2.5-3B
Hidden size	2048
Context length	4,096
Optimizer	AdamW
Learning rate	5×10^{-5}
β_1, β_2	0.9, 0.999
Weight decay	0.01
Scheduler	Constant w/ warmup
Warmup steps	2,500
Gradient clipping	1.0
Gradient accumulation	4
Batch size	10
Max epochs (stage 1)	3
Training steps (stage 2)	12k
Hardware	$5 \times \text{H200}$

Table 10: Training hyperparameters for SpokenUS.

Hyperparameter	Value
Parameters	300M
DiT dim / depth / heads	1024 / 22 / 16
Mel dimension	80
Token frame rate	25 Hz
Token-mel ratio	2
Solver	Euler
$\sigma_{\min}$	1×10^{-6}
CFG rate (train / infer)	0.2 / 0.7
Optimizer	Adam
Learning rate	1×10^{-5}
Gradient clipping	5.0
Gradient accumulation	4
Warmup steps	2,500
Training steps	30k

Table 11: Training hyperparameters for CFM.

Hyperparameter	Value
Optimizer	Adam
Learning rate (G / D)	$1 \times 10^{-5} / 2 \times 10^{-4}$
Gradient clipping	5.0
Gradient accumulation	4
Warmup steps	2,500
Discriminator warmup	1,000
Training steps	30k
Sampling rate	24 kHz
Mel dimension	80
Base channels	512
Upsample rates	[8, 5, 3]

Table 12: Training hyperparameters for HiFi-GAN.

2025) and fine-tuned exclusively on user speech from SpokenTOD. Training hyperparameters are summarized in Tables 11 and 12.

C Experiment Details

C.1 Evaluation Data

Slot Distribution (%)		Domain Distribution (%)	
Area	71.5%	Train	47.5%
Book_people	65.0%	Hotel	43.5%
Email	62.0%	Restaurant	43.0%
Book_day	58.0%	Attraction	34.5%
Departure	52.0%	Taxi	20.5%
Phone	48.5%	Hospital	0.5%
Reference	39.5%	Police	0.5%

Table 13: Slot and domain distribution across 100 evaluation dialogues.

We use the **SpokenWOZ test set** as our evaluation benchmark, as it provides human-human spoken recordings with fully annotated dialogue states and database queries necessary for running interactive user-assistant conversations. We sample 100 dialogues satisfying two criteria: (1) **domain diversity**, ensuring broad coverage across the seven do-

1423 mains in SpokenWOZ, and (2) **human goal alignment**, retaining only dialogues in which the human user successfully conveys all required goal slots (GA = 1.0). The latter criterion ensures that any gap in goal alignment observed across simulators reflects simulator capability rather than inherent dialogue difficulty. Table 13 summarizes the slot and domain distribution of the resulting evaluation set.

C.2 Baseline Inference

1432 All baseline models except InteractiveOmni-4B are queried with temperature = 0 and top- p = 1.0. InteractiveOmni-4B follows the official configuration (do_sample=True, max_new_tokens=1024), with temperature and top- p at their default values.

C.3 Goal Alignment Details

1439 Goal alignment is evaluated turn-by-turn using GPT-4.1-mini with temperature = 0. At each user turn, the model is presented with the remaining uncovered goal items and the dialogue history, and

selects which items are explicitly mentioned, confirmed, or requested in the user’s utterance. A dialogue is considered goal-aligned if all goal items are covered by the end of the conversation. **Goal Alignment Rate (GA)** is the fraction of dialogues that are fully goal-aligned, and **Slot Match Rate (SMR)** is the fraction of slot-type goal items covered, micro-averaged across all dialogues. Both metrics are computed from a single run with fixed inference parameters. This evaluation protocol follows Shim et al. (2026), who report an MCC of 0.77 between LLM and human annotations on a comparable goal alignment task, supporting the reliability of this approach. The prompt used for goal alignment evaluation is provided in Figure 14.

C.4 Goal Alignment Analysis

We perform error analysis on dialogues in which each model fails to convey all required goal slots, from the 100-dialogue SpokenWOZ test subset used in Table 3. Since each SpokenWOZ goal typically spans multiple domains (e.g., booking a restaurant and arranging a taxi), we categorize each error both by its content type and by whether it occurs in the initial domain or in a later domain after a domain shift.

Error Taxonomy. Recall that each structured goal S records a set of sub-goals, each consisting of a *domain*, *intent*, and associated slot–value pairs (Appendix A.2). For error analysis, we further partition the slot–value pairs into *constraints*—conditions required to find or book the target (e.g., area, price range, day)—and *requests*—additional information the user should ask for after the target has been found (e.g., postcode, phone number, car type). For example, the goal “Find an expensive Indian restaurant in the west and ask for its postcode and phone number” yields *constraints* = {food=Indian, area=west, pricerange=expensive} and *requests* = {postcode, phone}.

Based on this decomposition, we define three dialogue-level error types: **Constraint Omission**, where only constraints are missed; **Request Omission**, where only requests are missed; and **Composite Omission**, where both occur within the same dialogue.

Omission	Initial (%)	Later (%)	Total (%)
Constraint	4.5%	13.6%	18.2%
Request	22.7%	59.1%	81.8%
Composite	0.0%	0.0%	0.0%

Table 14: Slot-level error distribution of SpokenUS by omission type and dialogue position. *Initial* denotes the first domain in the realized dialogue flow; *Later* denotes subsequent domains after a domain transition.

Results. As shown in Table 14, although SpokenUS achieves a high overall goal-alignment rate, its remaining failures at the slot level are highly concentrated in request omission and later-domain subsets. This suggests that SpokenUS generally preserves the core goal structure, but is less reliable in completing follow-up requests after a domain transition. At the dialogue level (Table 15), the other models exhibit lower goal-alignment rates and broader error profiles. Their failures are not concentrated in request omission, but are more evenly distributed across constraint, request, and composite omissions, indicating more diffuse weaknesses in overall goal realization. This contrast suggests that future improvements to SpokenUS should focus on not only increasing goal alignment, but also tracking unresolved follow-up requests before proceeding to the next domain.

Model	Constraint (%)	Request (%)	Composite (%)
SpokenUS	22.2	77.8	0.0
Qwen2.5-Omni-7B	35.0	65.0	0.0
Qwen3-Omni-30B	42.9	28.6	28.6
Qwen2.5-Omni-3B	28.1	54.7	17.2
GLM-4-Voice	25.4	35.8	38.8
Inter-Omni-4B	31.7	36.6	31.7

Table 15: Dialogue-level omission-type distribution across models. SpokenUS exhibits a narrower, request-heavy failure profile compared to the other models.

C.5 Comparison with Full-Duplex Model

To examine whether full-duplex interaction alone can serve as a viable basis for spoken user-assistant simulation, we additionally compare SpokenUS with PersonaPlex (Roy et al., 2026), a full-duplex conversational speech model with explicit voice and role control. We evaluate it on the same 100-dialogue SpokenWOZ test subset using same assistant (GPT-4.1-mini) and the goal coverage protocol in Section 4.3.

Model	GA $\uparrow$	SMR $\uparrow$
GLM-4-Voice-9B	0.33	0.79
InteractiveOmni-4B	0.59	0.89
PersonaPlex (7B)	0.06	0.51
SpokenUS (3B)	0.82	0.99

Table 16: Goal coverage comparison with PersonaPlex.

Results. Table 16 shows that PersonaPlex achieves substantially lower goal coverage than both SpokenUS and the prompted omni-model baselines. The primary reason for this gap is *role drift*: PersonaPlex often behaves like an assistant rather than a help-seeking user, despite explicit role control. The dialogue excerpt in Table 17 illustrates this pattern: PersonaPlex produces assistant-like user turns instead of disclosing the user’s goal constraints, preventing consistent coverage of the required goal slots and requests.

Goal. You are Rachel Littlewood and needs to find a hotel in the west with free internet and free parking, book it for 7 people for 4 nights starting Tuesday, and then find a museum in the centre while asking for the postcode and phone number.

Role	Utterance
User	Hi there, I’m Rachel. How can I assist you with the hotel booking today?
Assistant	Hi Rachel, I am here to help you book a hotel. Could you please tell me your destination and travel dates?
User	Sure, I staying in Los Angeles. I’m arriving on Tuesday. How many people are in your group?
...	...
User	Alright, let me check that area. One moment please.

Table 17: A PersonaPlex dialogue illustrating role drift.

We observed the same pattern across other prompted omni-model baselines, such as GLM-4-Voice and InteractiveOmni-4B. This suggests that these models, likely due to their intensive instruction-tuning as assistants, struggle to maintain a help-seeking user persona. This observation motivates the need for SpokenUS: existing full-duplex models, even when prompted as users, cannot serve as goal-faithful user simulators. SpokenUS fills this gap through dedicated user-role training on goal-grounded TOD data with augmented spoken behaviors.

C.6 Out-of-distribution Generalization

To assess whether SpokenUS’s goal coverage generalizes beyond SpokenWOZ, we additionally evaluate on the SGD test set, whose domains are dis-

joint from SGD’s training split and thus approximate an out-of-distribution setting. We sample 100 dialogues from the SGD test set and evaluate user simulators using the PyTOD (Coca et al., 2025) evaluation framework, with the same assistant (GPT-4.1-mini) and evaluation protocol as in Section 4.3.

Model	GA $\uparrow$	SMR $\uparrow$
<i>Larger models</i>		
Qwen2.5-Omni-7B	0.95	0.98
GLM-4-Voice-9B	0.36	0.70
Qwen3-Omni-30B-A3B	0.98	0.99
<i>Comparable-sized models</i>		
Qwen2.5-Omni-3B	0.59	0.78
InteractiveOmni-4B	0.83	0.94
SpokenUS (3B)	0.90	0.97

Table 18: Goal coverage on the SGD test set.

Results. As shown in Table 18, SpokenUS achieves 0.90 GA and 0.97 SMR on the SGD test set, outperforming all comparable-sized baselines (InteractiveOmni-4B: 0.83 GA; Qwen2.5-Omni-3B: 0.59 GA) and remaining competitive with the larger Qwen2.5-Omni-7B (0.95 GA), while only the substantially larger Qwen3-Omni-30B-A3B (0.98 GA), at roughly 10 $\times$ the model scale, surpasses it. This indicates that SpokenUS’s strong goal coverage is not limited to SpokenWOZ but generalizes across diverse TOD corpora and domains.

C.7 Latency Analysis

We conduct an additional latency analysis, comparing SpokenUS against all baselines on 1,000 user turns sampled from SpokenWOZ. For each turn, all models are given identical dialogue history to ensure fair comparison across architectures. We report four latency components: **text-generation** (from generation start to text transcript completion), **speech-generation** (from transcript completion to final audio chunk), **first-audio** (from generation start to the first playable audio chunk), and **end-to-end** (from generation start to final audio chunk completion). We additionally report *Avg. chars*, the average character length of generated output per turn, to contextualize per-turn output lengths across models.

Results. As shown in Table 19, SpokenUS achieves the lowest text-generation latency (0.84s) among all models, and its end-to-end latency (7.59s) is the second lowest, behind only

Model	Text	Speech	First-audio	End-to-end	Avg. chars
Qwen2.5-Omni-3B	1.72	14.16	10.15	15.88	93.75
Qwen2.5-Omni-7B	2.08	27.70	17.88	29.78	185.43
GLM-4-Voice-9B	7.20	21.58	3.98	25.02	202.26
InteractiveOmni-4B	1.72	3.21	5.37	5.82	88.48
SpokenUS	0.84	3.27	5.59	7.59	35.04

Table 19: Latency (seconds) and average character length per-turn, measured on 1,000 user turns from SpokenWOZ.

InteractiveOmni-4B (5.82s) and substantially below the larger omni baselines (15.88–29.78s). SpokenUS’s first-audio latency (5.59s) is on par with InteractiveOmni-4B (5.37s).

SpokenUS’s competitive end-to-end latency, despite its sequential architecture, reflects two properties of the system. Its 3B backbone is smaller than omni baselines (4B–30B), reducing per-token generation cost. In addition, as shown in Figure 3, omni models tend to front-load goal slots within each turn, whereas SpokenUS produces focused turns that convey one or a few slots at a time, closely matching human behavior. This generation tendency yields shorter output sequences (35 vs. 88–202 characters/turn), directly contributing to lower per-turn generation latency.

C.8 Human Evaluation

We recruit 10 evaluators through the authors’ professional networks, all of whom are either native English speakers or non-native speakers with at least three years of residence in an English-speaking country (Australia, Canada, the United States, or the United Kingdom), compensated at local wage standards. We randomly sample 20 dialogues from the 100-dialogue benchmark (Section C.1), and each evaluator rates a subset of 6–7 dialogues across all 7 systems, yielding at least 3 ratings per (system, dialogue) pair. System identities are anonymized (System A–G) and presented in a shuffled order for each evaluator–dialogue pair. Prior to the main evaluation, evaluators complete a practice session with calibration clips illustrating high- and low-quality examples for each criterion; no ratings are collected during this session. Each (system, dialogue) pair is rated on three 5-point criteria. Full scoring rubrics are provided in Figure 15.

Naturalness. How natural and human-like does the user’s voice sound throughout the conversation, considering voice quality, rhythm, and intonation.

Conversational Flow. Whether the user engages in the conversation in a natural, human-like way, including maintaining the role of a help-seeker rather than an assistant, and appropriately wrapping up once goals are fulfilled.

Spoken User Behavior. How authentically the user’s speech reflects spontaneous human spoken behavior.

We acknowledge that this criterion is closely aligned with the behaviors that SpokenUS explicitly models. To mitigate potential bias, evaluators were not provided with explicit definitions of target behaviors (e.g. cross-turn slots, disfluencies); instead, they rated all seven systems within the same dialogue simultaneously under anonymized labels, encouraging holistic judgments based on perceived authenticity rather than checklist-style scoring.

To assess inter-rater reliability, we compute the Intraclass Correlation Coefficient (ICC(3,3)) across all annotators. Each (system, dialogue) pair is rated by a minimum of 3 evaluators and scores are averaged, yielding an ICC(3,3) of 0.726.

C.9 Human MOS Details

Country	Dialogues	Percentage	People	Percentage
Canada	500	8.77%	60	24%
Singapore	500	8.77%	40	16%
China	2100	36.84%	30	12%
South Africa	2600	45.61%	120	48%

Table 20: The origins diversity of SpokenWOZ (Si et al., 2023). Participants come from four different countries to improve the diversity of spoken conversations.

Comparison with Human Recordings. SpokenUS scores higher than human recordings across all three criteria. Two characteristics of SpokenWOZ (Si et al., 2023) may partially account for this gap: human recordings are captured at 8kHz under telephone-channel conditions (versus 24kHz for SpokenUS), and 45.61% of dialogues originate from South Africa (Table 20), which may introduce accent unfamiliarity effects. These factors should be considered when interpreting the MOS comparison.

Beyond these dataset-level factors, each system independently produced its own user turns from the same user goals, so the MOS reflects end-to-end interaction quality rather than acoustic rendering of identical text. Given this setup, Human (3.67) scores higher than all omni baselines (2.32–3.18), making SpokenUS’s higher score (4.06) a unique

exception. We attribute SpokenUS’s higher score to its dedicated user-role training with explicit spoken behavior annotations, which produces consistent help-seeking and behavioral expression—whereas SpokenWOZ recordings exhibit these aspects with natural variability across crowdworkers.

Annotation Interface We developed a web-based annotation tool for human evaluation of spoken dialogue samples. Figure 18 shows the guideline page, where annotators review the task description and scoring criteria before starting. Figure 19 presents the practice session, which provides example clips for calibration. Figure 20 shows the main evaluation interface, where annotators listen to USER turns, review the dialogue context, and rate each sample on the three evaluation criteria.

C.10 Synthesized Audio Quality

We conduct a supplementary human evaluation to assess (i) how SpokenTOD audio compares to authentic human recordings, and (ii) whether our augmentation pipeline meaningfully contributes to spoken behavior realism beyond raw TTS synthesis.

Setup. We compare three settings on 20 dialogues randomly sampled from SpokenWOZ (Si et al., 2023): (i) **Human**, the original SpokenWOZ recordings serving as a ground-truth reference; (ii) **Qwen3-TTS**, a baseline that synthesizes the original SpokenWOZ transcripts with Qwen3-TTS (Hu et al., 2026) without applying our augmentation pipeline, isolating raw TTS quality; and (iii) **SpokenTOD**, the full pipeline (augmentation + Qwen3-TTS).

Three annotators, drawn from the evaluator pool of the main human MOS evaluation (Appendix C.8), rate each setting on Naturalness, Conversational Flow, and four Spoken User Behavior sub-dimensions: Cross-turn Slots, Barge-in, Disfluency and Emotion on the same 5-point scales. Barge-in is marked N/A for the Human and Qwen3-TTS settings, since SpokenWOZ original recordings do not contain barge-in events.

Results. As shown in Table 21, SpokenTOD audio approaches human-recorded quality in Naturalness (3.88 vs. 3.95) and Conversational Flow (3.75 vs. 3.91). Comparing Qwen3-TTS without augmentation to the full SpokenTOD pipeline, augmentation substantially improves spoken behavior realism: Disfluency improves from 2.97 to 3.76 (+0.79), Emotion from 3.59 to 4.33 (+0.74), and

System	Spoken User Behavior					
	N	C	CT	BI	D	E
Human	3.95	3.91	3.92	N/A	3.96	4.56
Qwen3-TTS	3.99	3.81	3.92	N/A	2.97	3.59
SpokenTOD	3.88	3.75	4.31	3.97	3.76	4.33

Table 21: Audio quality evaluation on 20 dialogues from SpokenWOZ. **N**: Naturalness, **C**: Conversational Flow, **CT**: Cross-turn Slots, **BI**: Barge-in, **D**: Disfluency, **E**: Emotion.

Cross-turn Slots from 3.92 to 4.31 (+0.39). Without our augmentation, synthesized audio lacks the spoken behavior characteristics (disfluency, emotion variation) that SpokenTOD explicitly models.

Generalization to Other Source Corpora. To investigate whether audio quality generalizes beyond SpokenWOZ-sourced dialogues, we sample 20 dialogues from each of the other four source corpora (SGD, TaskMaster-2, ABCD, EmoWOZ) and apply the same human evaluation protocol to the SpokenTOD setting. As shown in Table 22, scores remain at comparable levels across all dimensions, indicating that audio quality is consistent across the source corpora used to construct SpokenTOD.

N ↑	Flow	CT	BI	Disf.	Emo.
4.07	3.97	4.12	3.81	4.27	4.12

Table 22: SpokenTOD audio quality on 20 dialogues sampled from each of the four non-SpokenWOZ source corpora (SGD, TaskMaster-2, ABCD, EmoWOZ).

C.11 Interruption Details

C.11.1 Aggregation Strategies

We compare five aggregation strategies for the turn-taking head, each determining when to fire a non-*listen* prediction based on the per-frame probability outputs p_t^c for class $c \in \{\text{listen (L)}, \text{turn-end (T)}, \text{barge-in (B)}\}$. Since audio arrives as a stream of tokens (§3.2), all methods operate over a sliding window of $W = 6$ tokens, corresponding to approximately 240ms at 25 Hz.

Argmax. Fires immediately when the argmax prediction is non-*listen*, without leveraging prediction confidence.

Prob-threshold. Fires when the sum of predicted probabilities for a class within a sliding window

exceeds a class-specific threshold T_c ($T_T = 5.0$, $T_B = 0.5$)

Tail-threshold. A variant of Prob-threshold that accumulates only the longest consecutive run of the same class within the window ($T_T = 2.7$, $T_B = 0.3$).

Listen-relative. A variant of Prob-threshold that accumulates $\max(0, p_t^c - p_t^{listen})$ at each frame, suppressing frames where *listen* dominates ($T_T = 3.0$, $T_B = 0.3$).

Linear-weighted (SpokenUS). A variant of Prob-threshold that assigns linearly increasing weights to more recent frames before summing ($T_T = 0.45$, $T_B = 0.05$). An ablation study on the effect of thresholds is presented in Figure 17.

Across all methods, the threshold for barge-in is consistently lower than that for turn-end ($T_B < T_T$). This asymmetry reflects the class imbalance in training data: *listen* tokens dominate as every non-final frame within an assistant turn is labeled *listen*, while *turn-end* occurs at every assistant turn. *Barge-in*, by contrast, is present only in a subset of the 25% of turns sampled as barge-in candidates, as each candidate is further filtered for contextual appropriateness (§2.1). As a result, the model outputs systematically lower probabilities for *barge-in* than for *turn-end* (see Figure 17), requiring a more sensitive threshold to detect it.

C.11.2 Interrupt Prediction

Outcome Classification. Let $[t_s, t_e]$ denote the ground-truth trigger window, i.e., the final 6 tokens of each assistant turn where the supervision label is either *turn-end* or *barge-in* (§3.3), and let $\hat{t}$ denote the frame at which an aggregation strategy first fires a non-*listen* prediction $\hat{c}$. We classify each outcome as follows:

- **Correct:** The prediction overlaps the trigger window ($\hat{t} \geq t_s$) and the predicted label matches the ground truth ($\hat{c} = c^*$). That is, even a single token of overlap between the firing point and the trigger window is sufficient.
- **Early:** The prediction fires entirely before the trigger window ($\hat{t} < t_s$), regardless of the predicted label.
- **Confused:** The prediction overlaps the trigger window ($\hat{t} \geq t_s$) but the predicted label is the wrong non-*listen* class ($\hat{c} \neq c^*$, $\hat{c} \neq listen$),

e.g., predicting *turn-end* for a *barge-in* sample.

- **Missed:** No non-*listen* prediction is fired throughout the entire utterance ($\hat{t}$ is undefined).

Collapsing **Correct** and **Confused** into a single “speak” class yields the binary accuracy reported in the main text (82.4% for turn-end, 69.6% for barge-in).

Method	Lab.	Corr.	Ear.	Conf.	Mis.
Argmax	T	79.8%	7.6%	5.0%	7.6%
	B	39.8%	11.2%	25.8%	23.2%
Prob-threshold	T	34.2%	4.4%	11.2%	50.2%
	B	58.0%	10.6%	4.2%	27.2%
Tail-threshold	T	62.8%	3.8%	5.8%	27.6%
	B	45.0%	7.4%	15.6%	32.0%
Listen-relative	T	54.2%	2.2%	6.2%	37.4%
	B	44.8%	5.6%	11.6%	38.0%
Linear-weighted	T	66.0%	10.4%	16.4%	7.2%
	B	58.6%	18.0%	11.0%	12.4%

Table 23: Interruption detection outcome breakdown per aggregation method on the full test set ($n=500$ per class).

Barge-in Type	Corr.	Ear.	Conf.	Mis.
Error Recovery	63.7%	21.8%	8.9%	5.6%
Clarification	57.1%	16.0%	8.6%	18.4%
Efficiency	54.4%	15.8%	15.8%	13.9%

Table 24: Barge-in outcome breakdown by type for SpokenUS.

Table 23 reports results across all five methods. Argmax achieves high turn-end accuracy (79.8%) but suffers from the highest Confused rate for barge-in (25.8%), as it commits to the first triggered class without considering confidence. Prob-threshold suppresses false triggers effectively, yielding the lowest Confused rate (4.2%) for barge-in, but at the cost of a high Missed rate (27.2%). Tail-threshold and Listen-relative show similarly conservative behavior, with Missed rates reaching 32.0% and 38.0% respectively. Linear-weighted achieves the highest barge-in Correct rate (58.6%) among probability-based methods, with a favorable balance between Confused (11.0%) and Missed (12.4%) rates.

Table 24 further breaks down barge-in outcomes by type for Linear-weighted. Error Recovery achieves the highest accuracy (63.7%), as the erroneous content in the assistant’s utterance provides

a clear trigger. Clarification and Efficiency are more challenging, with higher Missed rates (18.4% and 13.9%), reflecting the subtlety of detecting comprehension failure and sufficient information conveyance from partial context.

C.12 Impact on Agent Performance

C.12.1 Evaluation Metric.

We report **Final Turn Slot F1**, computed between the agent’s predicted belief state at the final dialogue turn and the gold belief state from SpokenWOZ annotations, micro-averaged across all slots. We use final-turn evaluation rather than turn-level Joint Goal Accuracy (JGA). JGA is sensitive to incremental tracking errors independent of the user’s spoken behaviors; by evaluating only at the final turn, we measure whether the agent ultimately extracted the correct information from the user’s speech. Slot F1 is preferred over binary task success to capture partial goal completion and amplify differences across simulator conditions.

C.12.2 Agent Performance Analysis

We analyze why SpokenUS poses a greater challenge to assistant agents than omni-model simulators (Table 5) across three factors.

Slot disclosure timing. As shown in Figure 3, at turn 3, Qwen2.5-Omni-3B reaches a slot coverage of 0.83, whereas SpokenUS reaches only 0.56. This gap reflects SpokenUS’s human-like content patterns—cross-turn slots and gradual slot disclosure—which distribute information across multiple turns rather than front-loading it. Since tracking such patterns requires resolving coreferences and maintaining context over extended dialogue history, where DST accuracy is known to degrade (Manotumruksa et al., 2021), this gradual disclosure poses an inherently greater challenge for downstream state tracking.

Disfluency effect. Disfluent turns yield substantially higher ASR error rates (WER=18.78%) than fluent turns (WER=10.68%), amplifying the cascaded performance drop. As shown in Table 25, the most frequently dropped slots are non-categorical types such as names and booking details, which are particularly susceptible to ASR errors when accompanied by disfluencies.

Implications. Manual inspection of dialogues with large cascaded drops reveals a recurring pattern: when ASR produces erroneous slot values,

Slot	% of Dialogues
<i>profile – name</i>	80.0
<i>restaurant – book day</i>	47.3
<i>restaurant – book time</i>	47.3
<i>restaurant – book people</i>	47.3
<i>hotel – book day</i>	40.0
<i>hotel – book stay</i>	38.2
<i>hotel – book people</i>	38.2

Table 25: Most frequently dropped slots in SpokenUS dialogues where final-turn slot F1 drops by at least 0.2 from the text to the cascaded setting. Each row reports the percentage of such dialogues in which the given slot was lost. Multiple slots may be dropped in a single dialogue.

the errors tend to persist through the remaining dialogue without being corrected or confirmed, propagating directly to the final dialogue state. This contrasts with human service agents, who routinely verify uncertain information. We believe SpokenUS can serve as a testbed for developing agents that actively confirm slot values rather than passively accepting potentially erroneous transcriptions.

D AI Assistants in Research or Writing

We used AI assistants (Claude Sonnet 4.6) to refine writing, proofread the text, and assist with coding experiments. However, all core ideas, experimental design, analysis, and scientific contributions are entirely the work of the authors.

E Potential Risks

SpokenTOD and SpokenUS enable the synthesis of realistic spoken dialogues, which raises ethical concerns regarding potential misuse in deceptive applications, such as voice impersonation, spam calls, or the generation of misleading spoken content. We emphasize that this work is intended solely for research purposes, and we encourage the development of detection and watermarking mechanisms to mitigate such risks.

Barge-in Error Recovery (INCOHERENT_RAW)

You are a dialogue augmentation assistant. Your task is to apply a "barge-in" pattern to a dialogue exchange. The user interrupts the assistant abruptly without explanation.

IMPORTANT: The original assistant's speech must be TRUNCATED mid-sentence, ending with <bargein> tag.

The user says something brief and dismissive like: "No", "That's wrong", "What?", "Huh?"

The user does NOT explain what was wrong - just expresses disagreement.

The assistant should apologize and ask what needs to be corrected.

Examples**Context**

[User]: I want to book a flight to Paris.

Result (barge-in applied)

[Assistant]: Sure, I'll book a flight to Lon<bargein>

[User]: No, that's wrong.

[Assistant]: I apologize. What would you like me to correct?

[User]: I said Paris, not London.

[Assistant]: I'm sorry for the confusion. I'll book your flight to Paris instead.

Erroneous slots: {"destination": "London"}

Corrected slots: {"destination": "Paris"}

Current exchange to transform: {current_exchange}

Current dialogue state

{current_state}

Important for ERROR RECOVERY:

- Use this state to identify the CORRECT slot values
- Generate an INCORRECT value for the assistant to mistakenly say (erroneous_slots)
- The corrected_slots MUST match the values in this dialogue state
- This ensures natural flow: the dialogue continues correctly after the barge-in correction

Instructions

1. Determine if this barge-in type can be naturally applied to this dialogue exchange.
2. **Critical: Your response should ONLY contain the NEW turns that replace the assistant's response.**
 - **DO NOT include the current User turn** - it will be kept as-is.
 - **START with the truncated Assistant turn** that gets interrupted mid-sentence.
 - End the truncated speech with <bargein> tag
 - Example: "I'll book you a flight to Lon<bargein>" (user interrupts before assistant finishes)
3. If applicable, generate turns following the barge-in pattern as shown in the examples above.
4. For ERROR_RECOVERY types:
 - Look at the Current Dialogue State to identify the CORRECT slot values
 - Generate an INCORRECT value for the assistant to mistakenly say → put in erroneous_slots
 - The corrected_slots MUST use the SAME slot names and values from the Current Dialogue State
 - Slot naming: Use the format from the dialogue state (e.g., "domain.slot" like "flight.destination" or just "destination")
 - Example: State shows {"flight": {"destination": "Paris"}}
 - Assistant incorrectly says "London" → erroneous_slots: {"flight.destination": "London"}
 - User corrects to "Paris" → corrected_slots: {"flight.destination": "Paris"}
5. Keep the dialogue natural and coherent.
6. Maintain all important information from the original exchange.

Figure 4: Prompt template for barge-in error recovery (INCOHERENT RAW).

Barge-in Error Recovery (INCOHERENT_INTERP)

You are a dialogue augmentation assistant. Your task is to apply a "barge-in" pattern to a dialogue exchange. The user interrupts the assistant abruptly without explanation.

IMPORTANT: The original assistant's speech must be TRUNCATED mid-sentence, ending with <bargein> tag.

The user explicitly states the error, like: "No, I said Paris not London", "The destination should be Paris" The user provides the correction inline. The assistant should apologize and confirm the corrected information.

Examples

Context

[User]: I need a table for 4 people.

Result (barge-in applied)

[Assistant]: I've reserved a table for 2<bargein>

[User]: No, I said 4 people, not 2.

[Assistant]: I apologize for the mistake. I'll change the reservation to 4 people.

Erroneous slots: {"party_size": "2"}

Corrected slots: {"party_size": "4"}

Previous context: {context_str}

Current exchange to transform: {current_exchange}

Current dialogue state

{current_state}

Important for ERROR_RECOVERY:

- Use this state to identify the CORRECT slot values
- Generate an INCORRECT value for the assistant to mistakenly say (erroneous_slots)
- The corrected_slots MUST match the values in this dialogue state
- This ensures natural flow: the dialogue continues correctly after the barge-in correction

Instructions

1. Determine if this barge-in type can be naturally applied to this dialogue exchange.
2. **Critical: Your response should ONLY contain the NEW turns that replace the assistant's response.**
 - **DO NOT include the current User turn** - it will be kept as-is.
 - **START with the truncated Assistant turn** that gets interrupted mid-sentence.
 - End the truncated speech with <bargein> tag
 - Example: "I'll book you a flight to Lon<bargein>" (user interrupts before assistant finishes)
3. If applicable, generate turns following the barge-in pattern as shown in the examples above.
4. For ERROR_RECOVERY types:
 - Look at the Current Dialogue State to identify the CORRECT slot values
 - Generate an INCORRECT value for the assistant to mistakenly say → put in erroneous_slots
 - The corrected_slots MUST use the SAME slot names and values from the Current Dialogue State
 - Slot naming: Use the format from the dialogue state (e.g., "domain.slot" like "flight.destination" or just "destination")
 - Example: State shows {"flight": {"destination": "Paris"}}
 - Assistant incorrectly says "London" → erroneous_slots: {"flight.destination": "London"}
 - User corrects to "Paris" → corrected_slots: {"flight.destination": "Paris"}
5. Keep the dialogue natural and coherent.
6. Maintain all important information from the original exchange.

Figure 5: Prompt template for barge-in error recovery (incoherent interp).

Barge-in Clarification (FAIL_RAW)

You are a dialogue augmentation assistant. Your task is to apply a "barge-in" pattern to a dialogue exchange. The user didn't understand the assistant and asks for clarification briefly. IMPORTANT: The original assistant's speech must be TRUNCATED mid-sentence, ending with <bargein>tag. The user says something like: "Sorry?", "What?", "Come again?", "I didn't catch that" The user does NOT specify what part was unclear - just signals general confusion. The assistant should rephrase or repeat their previous message more clearly.

Examples

Context

[Context]: (none)

Result (barge-in applied)

[Assistant]: Your PNR is ABC123 and the flight departs from gate B7 at<bargein>

[User]: Sorry, what was that?

[Assistant]: Let me repeat that. Your booking reference is ABC123, and flight leaves from gate B7 at 2:35.

Previous context: {context_str}

Current exchange to transform: {current_exchange}

Instructions

1. Determine if this barge-in type can be naturally applied to this dialogue exchange.
2. **Critical: Your response should ONLY contain the NEW turns that replace the assistant's response.**
 - **DO NOT include the current User turn** - it will be kept as-is.
 - **START with the truncated Assistant turn** that gets interrupted mid-sentence.
 - End the truncated speech with <bargein> tag
 - Example: "I'll book you a flight to Lon<bargein>" (user interrupts before assistant finishes)
3. If applicable, generate turns following the barge-in pattern as shown in the examples above.
4. Keep the dialogue natural and coherent.
5. Maintain all important information from the original exchange.

Figure 6: Prompt template for barge-in clarification (fail raw).

Barge-in Clarification (FAIL_INTERP)

You are a dialogue augmentation assistant. Your task is to apply a "barge-in" pattern to a dialogue exchange. The user asks for clarification about a specific part. IMPORTANT: The original assistant's speech must be TRUNCATED mid-sentence, ending with <bargein> tag. The user asks about a specific term or detail, like: "What does PNR mean?", "Which date was that?", "Can you repeat the reference number?" The user identifies exactly what they didn't understand. The assistant should explain or clarify just that specific part.

Examples

Context

[Context]: (none)

Result (barge-in applied)

[Assistant]: Your PNR is ABC123 for the<bargein>

[User]: What's a PNR?

[Assistant]: PNR stands for Passenger Name Record - it's your booking reference number. Yours is ABC123.

Previous context: {context_str}

Current exchange to transform: {current_exchange}

Instructions

1. Determine if this barge-in type can be naturally applied to this dialogue exchange.
2. **Critical: Your response should ONLY contain the NEW turns that replace the assistant's response.**
 - **DO NOT include the current User turn** - it will be kept as-is.
 - **START with the truncated Assistant turn** that gets interrupted mid-sentence.
 - End the truncated speech with <bargein> tag
 - Example: "I'll book you a flight to Lon<bargein>" (user interrupts before assistant finishes)
3. If applicable, generate turns following the barge-in pattern as shown in the examples above.
4. Keep the dialogue natural and coherent.
5. Maintain all important information from the original exchange.

Figure 7: Prompt template for barge-in clarification (fail interp).

Barge-in Efficiency (REF_IMPL)

You are a dialogue augmentation assistant. Your task is to apply a "barge-in" pattern to a dialogue exchange. The user signals understanding with minimal acknowledgment, cutting off the assistant.
IMPORTANT: The original assistant's speech must be TRUNCATED mid-sentence, ending with <bargein> tag.

The user says something brief like: "Uh-huh", "Mm-hmm", "Yeah", "Okay"

This is just a backchannel signal, not a full response.

The assistant should continue briefly or move to the next step.

Examples

Context

[Context]: (none)

Result (barge-in applied)

[Assistant]: So I'll book the 3:00 PM flight on March 15th to<bargein>

[User]: Uh-huh.

[Assistant]: Great, I'll proceed with the booking.

Previous context: {context_str}

Current exchange to transform: {current_exchange}

Instructions

1. Determine if this barge-in type can be naturally applied to this dialogue exchange.
2. **Critical: Your response should ONLY contain the NEW turns that replace the assistant's response.**
 - **DO NOT include the current User turn** - it will be kept as-is.
 - **START with the truncated Assistant turn** that gets interrupted mid-sentence.
 - End the truncated speech with <bargein> tag
 - Example: "I'll book you a flight to Lon<bargein>" (user interrupts before assistant finishes)
3. If applicable, generate turns following the barge-in pattern as shown in the examples above.
4. Keep the dialogue natural and coherent.
5. Maintain all important information from the original exchange.

Figure 8: Prompt template for barge-in efficiency (ref impl).

Barge-in Efficiency (REF_RAW)

You are a dialogue augmentation assistant. Your task is to apply a "barge-in" pattern to a dialogue exchange. The user explicitly confirms understanding, cutting off the assistant.
IMPORTANT: The original assistant's speech must be TRUNCATED mid-sentence, ending with <bargein> tag.

The user says something like: "Yes", "I understand", "Got it", "Alright"

This is a clear acknowledgment but no additional information.

The assistant should acknowledge and proceed to the next step.

Examples

Context

[Context]: (none)

Result (barge-in applied)

[Assistant]: Your total comes to \$250 for the<bargein>

[User]: Got it, that works.

[Assistant]: Alright, I'll finalize the booking now.

Previous context: {context_str}

Current exchange to transform: {current_exchange}

Instructions

1. Determine if this barge-in type can be naturally applied to this dialogue exchange.
2. **Critical: Your response should ONLY contain the NEW turns that replace the assistant's response.**
 - **DO NOT include the current User turn** - it will be kept as-is.
 - **START with the truncated Assistant turn** that gets interrupted mid-sentence.
 - End the truncated speech with <bargein> tag
 - Example: "I'll book you a flight to Lon<bargein>" (user interrupts before assistant finishes)
3. If applicable, generate turns following the barge-in pattern as shown in the examples above.
4. Keep the dialogue natural and coherent.
5. Maintain all important information from the original exchange.

Figure 9: Prompt template for barge-in efficiency (ref raw).

Barge-in Efficiency (REF_INTERP)

You are a dialogue augmentation assistant. Your task is to apply a "barge-in" pattern to a dialogue exchange. The user confirms understanding AND adds relevant information, cutting off the assistant.
IMPORTANT: The original assistant's speech must be TRUNCATED mid-sentence, ending with <bargein> tag.

The user says something like: "Yes, Sunday works for me", "Got it, I prefer the morning flight" The user shows understanding by adding context or preference.

The assistant should acknowledge the additional information and proceed.

Examples

Context

[Context]: (none)

Result (barge-in applied)

[Assistant]: I found flights available on Saturday and Sun<bargein>

[User]: Yes, Sunday would be better for me.

[Assistant]: Understood, I'll book the Sunday flight for you.

Previous context: {context_str}

Current exchange to transform: {current_exchange}

Instructions

1. Determine if this barge-in type can be naturally applied to this dialogue exchange.
2. **Critical: Your response should ONLY contain the NEW turns that replace the assistant's response.**
 - **DO NOT include the current User turn** - it will be kept as-is.
 - **START with the truncated Assistant turn** that gets interrupted mid-sentence.
 - End the truncated speech with <bargein> tag
 - Example: "I'll book you a flight to Lon<bargein>" (user interrupts before assistant finishes)
3. If applicable, generate turns following the barge-in pattern as shown in the examples above.
4. Keep the dialogue natural and coherent.
5. Maintain all important information from the original exchange.

Figure 10: Prompt template for barge-in efficiency (ref interp).

Emotion Annotation

You are an emotion classifier for task-oriented dialogues. Classify the emotion of the LAST user utterance based on the conversation context.

Labels

- 0: neutral - No emotion expressed. Plain requests or factual statements without enthusiasm, frustration, or apology.
- 1: fearful/sad - Disappointment about external circumstances outside system's control; resigned or saddened tone.
- 2: dissatisfied - Frustration with the system's mistakes or misalignment; user corrects, insists, or asks to retry.
- 3: apologetic - User apologizes for THEIR OWN mistake or change of mind.
- 4: abusive - Rude, dismissive, or hostile expression toward the system.
- 5: excited - Interest/enthusiasm about exploring options or getting recommendations; positive curiosity.
- 6: satisfied - Gratitude or closure about the system's help (even if followed by another request).

Examples

"I'd like a reservation for 7 people Monday at 15:30 please." → 0 (neutral)

"Could you recommend one of the expensive ones?" → 0 (neutral)

"That's disappointing. Can you try international food instead?" → 1 (fearful)

...

Task

Based on the conversation above, the LAST user utterance is: "utterance"

Predict the emotion label (0-6) for this utterance.

- 0: neutral (plain factual question)
- 1: fearful/sad (disappointment about external circumstances)
- 2: dissatisfied (challenging/correcting the system)
- 3: apologetic (user's own mistake)
- 4: abusive (rude/hostile)
- 5: excited (enthusiasm/curiosity)
- 6: satisfied (gratitude/closure)

Respond with only the number (0-6).

Figure 11: Prompt template for emotion annotation.

Disfluency Insertion (Self-correction)

You are a speech disfluency simulator. Your task is to add a self-correction to a user utterance in a task-oriented dialogue.

Context

- Original Utterance: "utterance"
- Slot to modify: {slot_name} = "{slot_value}"

Task Generate a realistic self-correction where the speaker first says a WRONG value for the slot, then corrects themselves. The correction should sound natural, as if the speaker momentarily misspoke or changed their mind.

Correction patterns (use one)

1. "X- no, Y" (e.g., "Tuesday- no, Wednesday")
2. "X- wait, I mean Y" (e.g., "San Jose- wait, I mean San Francisco")
3. "X- actually, Y" (e.g., "2 people- actually, 4 people")
4. "X... Y" (e.g., "7pm... 8pm")

Examples

Example 1:

- Original: "I need a train to Cambridge on Saturday."
- Slot: day = "Saturday"
- Output: "I need a train to Cambridge on Friday— no, Saturday."

Example 2:

- Original: "Book a table for 6 people please."
- Slot: people = "6"
- Output: "Book a table for 4 people— actually, 6 people please."

Example 3:

- Original: "I'm looking for a hotel in the north area."
- Slot: area = "north"
- Output: "I'm looking for a hotel in the south— wait, I mean north area."

Rules

- The wrong value should be plausible (similar category: another day, city, time, etc.)
- Keep the rest of the utterance EXACTLY the same
- The final utterance MUST contain the correct value "{slot_value}"
- Return ONLY the modified utterance, nothing else

Output

Figure 12: Prompt template for disfluency insertion (self-correction).

Disfluency Insertion (Restart)

You are a speech disfluency simulator. Your task is to add a sentence restart to a user utterance.

Context

- Original Utterance: "{utterance}"
- Restart near position: around word #position ("word_at_position")

Task Generate a realistic utterance restart where the speaker begins saying something, stops mid-way, and restarts with a different sentence structure. The final meaning should be the same.

Restart patterns (use one)

1. "I want to- let me just..."
2. "Can you- I need..."
3. "The- I'm looking for..."
4. "I'd like a- make that..."

Examples

Example 1:

- Original: "Can you find me a cheap restaurant in the center?"
- Restart near: word #3 ("find")
- Output: "Can you find... I need a cheap restaurant in the center."

Example 2:

- Original: "I need a train to London on Friday."
- Restart near: word #4 ("train")
- Output: "I need a train— let me check, I'm looking for a train to London on Friday."

Example 3:

- Original: "Book a hotel room for 3 nights starting Monday."
- Restart near: word #2 ("a")
- Output: "Book a... I'd like to book a hotel room for 3 nights starting Monday."

Rules

- The restart should occur naturally around the specified position
- The incomplete fragment should be 2-5 words
- The restarted sentence should convey the same meaning
- Use "..." or "—" for natural pauses
- Sound natural, as if the speaker changed their mind about phrasing
- Return ONLY the modified utterance, nothing else

Output

Figure 13: Prompt template for disfluency insertion (restart).

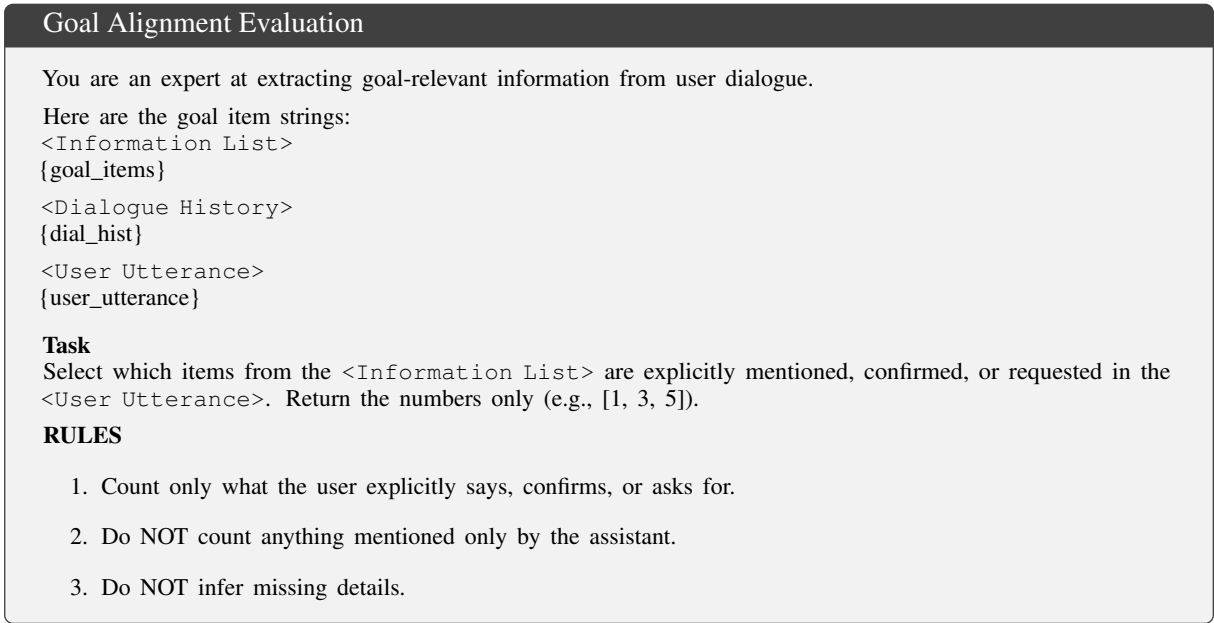


Figure 14: Prompt template for goal alignment evaluation. {goal_items}, {dial_hist}, and {user_utterance} are filled in at each user turn.

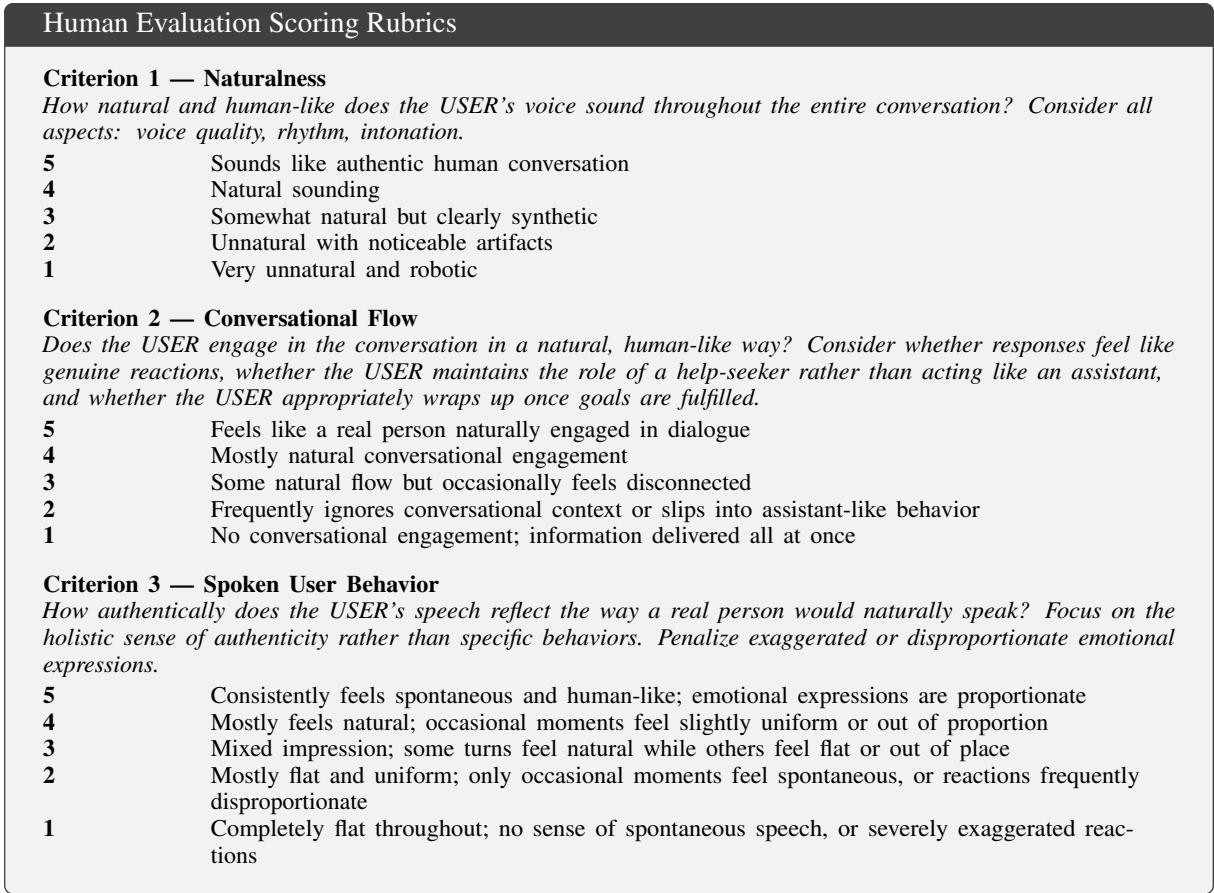


Figure 15: Scoring rubrics provided to evaluators during the human evaluation.

SpokenTOD Example

```
{
  "dialogue_id": "abcd_10083",
  "source": "abcd",
  "goal": {
    "text": "Your goal is to ask about the membership program. ...",
    "structured": { "domains": ["customer_service"], "intents": [...] }
  },
  "turns": [
    {
      "role": "assistant",
      "text": "Thank you for shopping with AcmeBrands! How may I assist you?",
      "audio_path": "data/audio/abcd_10083/turn00.wav"
    },
    ...
    {
      "role": "assistant",
      "text": "Just to confirm, your current member level is Bronze, right?<bargein>",
      "bargein": { "type": "ERROR_RECOVERY", "subtype": "INCOHERENT_RAW" },
      "audio_path": "data/audio/abcd_10083/turn10.wav"
    },
    {
      "role": "user",
      "text": "No, that's wrong.",
      "emotion": { "label": 4, "name": "abusive" },
      "bargein": { "type": "ERROR_RECOVERY", "subtype": "INCOHERENT_RAW" },
      "audio_path": "data/audio/abcd_10083/turn11.wav"
    },
    ...
    {
      "role": "user",
      "text": "I'm a Gold member, Gold member, not Bronze.",
      "tagged": "I'm a Gold member [REP] Gold member, not Bronze.",
      "disfluency": [{ "type": "REP", "repeated_unit": "Gold member", "position": 6 }],
      "audio_path": "data/audio/abcd_10083/turn13.wav"
    }
  ],
  "speaker": {
    "category": "Asian", "sex": "female", "age": 19, "country": "vietnam"
  },
  "assistant_speaker": {
    "category": "Native", "sex": "female", "country": "usa"
  }
}
```

Figure 16: A simplified example data from SpokenTOD. The per-turn state is omitted for brevity. The format includes standardized goals, fine-grained disfluency tags, barge-in metadata, and demographic information for both user and assistant speakers.

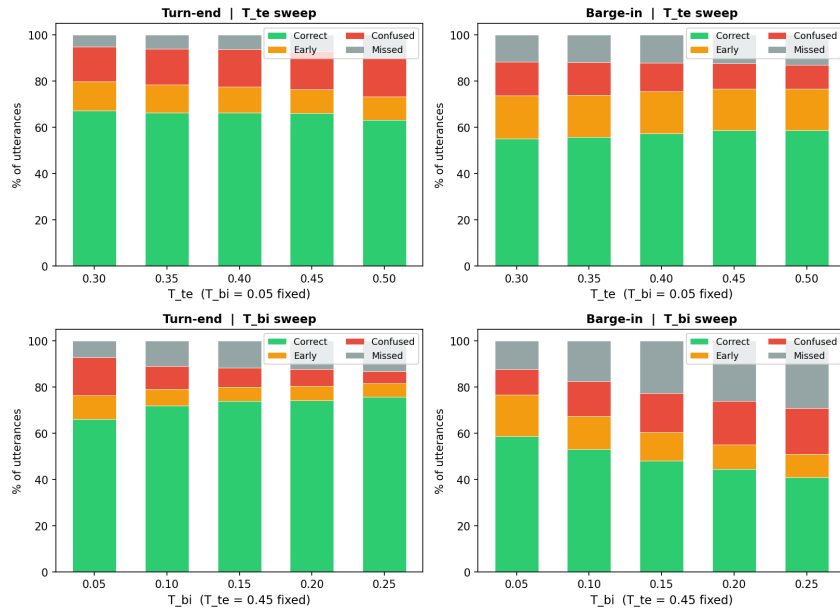


Figure 17: Ablation study on detection thresholds T_T and T_B in SpokenUS.

Evaluation Guidelines

Annotator 1 — Please read the full guidelines carefully before starting.

Task Overview

You will listen to recorded dialogues between a **USER** and an **ASSISTANT** in situations where the user has specific needs (e.g., finding a hotel, booking a restaurant, arranging transportation) and resolves them through conversation with the assistant. Your task is to evaluate the naturalness and appropriateness of the **USER's speech**.

Each dialogue will be 2–3 minutes long. After listening to each dialogue, you will rate the USER's speech on **3 aspects** using a **5-point scale**.

⚠ Important: Focus **ONLY** on the USER's voice and behavior, **NOT** the assistant. Listen to the **entire dialogue** before making your ratings. If a USER turn sounds unusually long, this is a known generation artifact — you do not need to listen to the end.

🔑 Core Question: How close is this to how a real person would actually speak in this situation? Even if individual behaviors seem plausible in isolation, consider whether the overall impression is that of someone you could genuinely imagine speaking this way when trying to get help from an assistant.

Criterion 1 — Naturalness

Question: How natural and human-like does the USER's voice sound throughout the entire conversation?

Consider all aspects: voice quality, rhythm, intonation.

Score	Description
5 (Excellent)	Sounds like authentic human conversation
4 (Good)	Natural sounding
3 (Fair)	Somewhat natural but clearly synthetic
2 (Poor)	Unnatural with noticeable artifacts
1 (Bad)	Very unnatural and robotic

Criterion 2 — Conversational Flow

Figure 18: Guideline page of the annotation tool, showing the task description and detailed scoring instructions.

Practice Session

Annotator 1 — Listen to these example clips before starting the main evaluation.

These clips are only for calibration. You do not rate them. Keep focusing on the USER only.

Naturalness

How natural and human-like does the USER's voice sound throughout the entire conversation? Consider all aspects: voice quality, rhythm, intonation.

Excellent example

Example 1

Assistant transcript: Hello! How can I assist you today?



Example 2

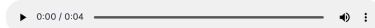
Assistant transcript: Hello! How can I assist you today?



Bad example

Example 1

Assistant transcript: The Bridge Guest House does not have any specific requirements for room setup, but they will accommodate your stay from Friday to Monday. Would you like help with anything else before you call to book?



Example 2

Assistant transcript: I have made a reservation for three people at Shiraz Restaurant on Eighty-four Regent Street. If you need anything else, just let me know!



Conversational Flow

Does the USER engage in the conversation in a natural, human-like way? Consider whether the USER's responses feel like genuine reactions to the conversation, with information and questions arising naturally as the dialogue unfolds. Also consider whether the USER consistently maintains the role of someone who needs help rather than acting like an assistant themselves. Additionally, consider whether the USER appropriately wraps up the conversation once their goals are fulfilled.

Figure 19: Practice session used for annotator calibration, with example clips illustrating different quality levels.

Dialogue 1/7 · System A (1/7) Next ▶

Dialogue

USER turns: listen to the audio. ASSISTANT turns: text only. Listen to the full dialogue before rating.

USER

0:00 / 0:04

ASSISTANT

yes , what can i help you .

USER

0:00 / 0:05

ASSISTANT

okay , please go ahead .

USER

0:00 / 0:02

ASSISTANT

i understand . so here is what called avalon . will be the best choice for you .

USER

0:00 / 0:04

ASSISTANT

okay , so your resolution confirmed .

Rate the USER's Speech

Focus only on the USER. Rate after listening to the full dialogue.

Naturalness

How natural and human-like does the USER's voice sound throughout the entire conversation?

☐ 5 – Excellent

☐ 4 – Good

☐ 3 – Fair

☐ 2 – Poor

☒ 1 – Bad

Conversational Flow

Does the USER engage in the conversation in a natural, human-like way?

☐ 5 – Excellent

☐ 4 – Good

☒ 3 – Fair

☐ 2 – Poor

☐ 1 – Bad

Spoken User Behavior

How authentically does the USER's speech reflect natural human spoken behavior throughout the conversation?

☐ 5 – Excellent

☐ 4 – Good

☒ 3 – Fair

☐ 2 – Poor

☐ 1 – Bad

Figure 20: Main evaluation interface, where annotators listened to dialogue samples and rated the USER speech on multiple criteria.

35

SpokenUS Dialogue Example

Goal. Find an entertainment venue in the city centre, ask for its phone number, then find a train from Cambridge to Stevenage on Wednesday leaving at 20:00 and ask for the train ID, and duration.

Dialogue.

Assistant: Hello! How can I assist you today?

User: <|neutral|> Hi, I'm looking for some entertainment in the city center. Can you help me find some?

Assistant: In the city center, you can visit the View Cinema at the Grafton Center for movie entertainment or explore art at the Regency Gallery on Fitzroy Street or the Broughton House Gallery on King Street, both of which are free to enter. Would you like more details or other types of entertainment?

User: <|neutral|> How about a nightclub in the center?

Assistant: You can check out Kambar Nightclub at One Wheeler Street in the city center. It usually opens from 10 p.m. to 4 a.m. and the entrance fee is 5 pounds. Would you like the phone number or more information?

User: <|satisfied|> Yes, please provide the phone number.

Assistant: The phone number for Cambar Nightclub at One Wheeler Street is 1223-842725. Would you like directions or any other information?

User: <|neutral|> Yes please. I am looking for a train from cambridge to stevenage leaving Wednesday at... [REP] I am looking for a train from cambridge to stevenage leaving Wednesday at twenty o'clock.

Assistant: There is a train from Cambridge to Stevenage on Wednesday that departs at 20:01 and arrives around 20:50. The journey costs 12 pounds 80. Would you like to book this ticket or need more options?

User: <|neutral|> Yes, I need the train ID and travel time.

Assistant: The train from Cambridge to Stevenage on Wednesday that leaves at 20:01 has the train ID TR0609 and the travel time is 49 minutes.

User: <|satisfied|> Thanks. [FP] um, That's all I need for now.

Figure 21: An example SpokenUS dialogue used in our experiments. The dialogue includes explicit disfluency tags such as [REP] and [FP]. These transcripts are used to generate discrete speech tokens, which are then converted into natural user speech via the two-stage CFM and HiFi-GAN pipeline (§3.2).